A compact HALEU reactor parked at JUNO: Standard-Model precision, sterile-neutrino wiggles, and axion-like particles

Internal report on notebooks 5–7 of the JUNO repository

(Dated: August 25, 2026)

This report summarises what I have done on the physics case for a compact high-assay low-enriched-uranium (HALEU) research reactor of $10\,\mathrm{MW_{th}}$ parked about 50 m from the JUNO central detector. Three studies are carried out within one analysis framework, built on the measured Daya Bay antineutrino spectra, JUNO's released detector response, and analytic Asimov profiling over joint Gaussian nuisance modes. First, elastic neutrino–electron scattering (EνES) at $\sim 3\times10^3$ events per day gives a $\sim 2\%$ determination of the low-energy weak mixing angle when analysed alone (where the measured flux shape is the limit), and $\sigma(\sin^2\theta_W) = 1.4\times10^{-3}$ (0.6%) when the 6×10^4 per day inverse-beta-decay (IBD) sample is used to anchor the flux in situ, with 8×10^{-4} within reach of plausible transfer systematics. JUNO's measured detector singles are included (4.7 Hz above 0.7 MeV, overwhelmingly external γ that stop at the 2.615 MeV ^{208}Tl line), and running the reactor off for as long as on measures them in situ, which leaves the answer intact even if they are thirty times larger than modelled. Second, for $\Delta m^2_{41}\sim 0.1$–$10\,\mathrm{eV}^2$ the sterile-neutrino oscillation length is metres, so JUNO's 33 m diameter turns the detector into a many-baseline experiment. A shape-only, vertex-binned wiggle search reaches $\sin^2 2\theta_{14}\approx 10^{-3}$ over 0.26–$1.1\,\mathrm{eV}^2$ and stays below 2×10^{-3} from 0.13 to $2.7\,\mathrm{eV}^2$, an order of magnitude beyond the reactor short-baseline programme. An integrated analysis of the same data with the same free flux shape has no sensitivity at all, so the finite detector is not an improvement on the conventional analysis but the only thing that makes a prior-free search possible; the full 3+1 formula is carried across standoffs of 50–1400 m. Third, the core's $\sim 5\times10^{18}\,\gamma$/s produce axion-like particles (ALPs) through Primakoff and Compton-like conversion; with an $855\,\mathrm{m}^2$ face and 22 m mean decay path, JUNO covers the "cosmological triangle" of the ALP–photon plane in full and reaches $g_{aee}\sim10^{-8}$ at MeV masses with the background included. Every cross section, decay rate and flux ingredient is validated against independent implementations or analytic benchmarks, and the systematics budgets and their limitations are stated explicitly.

CONTENTS

I. COMMON FRAMEWORK

All three studies share one set of inputs and one statistical machinery, so I describe them once here.

a. Source and fuel evolution. A HALEU core of thermal power P with fission fractions $f_i(\beta)$ ($i = {}^{235}$U, ^{238}U, ^{239}Pu, ^{241}Pu; $\sum_i f_i = 1$) at burnup position $\beta \in [0, 1]$ in its cycle has fission rate and antineutrino flux

$$R_f(\beta) = \frac{P}{\sum_i f_i(\beta)\, E_i}, \qquad \frac{d\Phi_\nu}{dE_\nu}(E_\nu, \beta) = \frac{R_f(\beta)}{4\pi L^2} \sum_i f_i(\beta)\, S_i(E_\nu), \tag{1}$$

with the thermal energies per fission $E_i = (202.36,\ 205.99,\ 211.12,\ 214.26)$ MeV [1] and S_i the per-fission antineutrino spectra. For $P = 10\,\mathrm{MW}$ this gives $R_f = 3.07 \times 10^{17}\,\mathrm{s}^{-1}$ at $\beta = 0.5$. The fresh composition is $f_{235} = 0.98$, $f_{238} = 0.02$, and plutonium grows in along the *Daya Bay* fuel-evolution trajectory [2]: with the measured ^{239}Pu and ^{241}Pu fission fractions $F_{239,241}(\beta_k)$ across Daya Bay's 20 burnup groups (β_k mapped linearly onto $[0, 1]$),

$$f_{239}(\beta) = F_{239}(\beta) - F_{239}(0), \qquad f_{241}(\beta) = F_{241}(\beta) - F_{241}(0), \qquad f_{238} = 0.02, \qquad f_{235} = 1 - f_{238} - f_{239} - f_{241}, \tag{2}$$

that is, the LEU *increments* of the Pu fractions are applied on top of the fresh HALEU composition at fixed ^{238}U. This is deliberately conservative: a 2%-^{238}U core breeds plutonium more slowly than the LEU cores the trajectory describes, so the true spectral drift is smaller than modelled. At the run-averaged $\beta = 0.5$ used throughout, $(f_{235}, f_{238}, f_{239}, f_{241}) = (0.901,\ 0.020,\ 0.063,\ 0.016)$; at end of cycle (0.818, 0.020, 0.125, 0.037). The fuel-evolution *systematic* is a single nuisance mode scaling the Pu increments by $(1 + \epsilon)$, $\sigma_\epsilon = 0.30$, whose spectral direction is

$$\frac{\partial}{\partial \epsilon} \sum_i f_i S_i = f_{239}\,(S_{239} - S_{235}) + f_{241}\,(S_{241} - S_{235}), \tag{3}$$

that is, fission share moved from ^{235}U into plutonium at fixed ^{238}U.

The per-fission spectra are the *measured* Daya Bay unfolded per-fission IBD yields for ^{235}U and ^{239}Pu [3], divided by the Vogel–Beacom cross section [4] wherever a flux rather than a yield is needed, together with the Huber–Mueller spectra [5, 6] for the percent-level ^{238}U and ^{241}Pu components (Section A). This flux is defined only above the IBD threshold of 1.806 MeV, so recoil spectra below $T \sim 1.5$ MeV are conservative.

b. Detector. I take JUNO's fiducial target to be $N_p = 1.44 \times 10^{33}$ free protons [7, 8], inside a fiducial radius of 16.5 m. That number is fixed by the delivered scintillator's hydrogen mass fraction of 12.01%, and the same composition fixes the electron count: $\mathrm{H/C} = 1.63$ by number, so one carbon comes with 1.63 free protons and 7.63 electrons and $N_e = 4.689 N_p = 6.76 \times 10^{33}$. Both target counts are derived from that one composition rather than quoted separately, because the joint fit of Section II measures the EνES/IBD rate ratio and N_e/N_p enters its answer directly. The response uses the released positron and electron non-linearities, an energy resolution $\sigma_E/E = \sqrt{a^2/E + b^2}$ with $a = 3.3\%$ and $b = 1.0\%$, a vertex resolution of $10\,\mathrm{cm}/\sqrt{E}$, a selection efficiency of 0.8, and a reactor duty cycle of 0.9.

A. Statistics

Every sensitivity in this report is an Asimov projection [9], and since the same construction is used in all four studies it is worth writing out once.

Let $\boldsymbol{\theta}$ be the physics parameters and $\boldsymbol{\xi}$ a set of nuisance parameters, each normalised so that it carries a unit Gaussian prior. The predicted mean in bin i is $\mu_i(\boldsymbol{\theta}, \boldsymbol{\xi})$, and the likelihood is the Poisson term for the data plus the prior term for the pulls,

$$\chi^2(\boldsymbol{\theta}, \boldsymbol{\xi}) = \sum_i \frac{\big[n_i - \mu_i(\boldsymbol{\theta}, \boldsymbol{\xi})\big]^2}{\mu_i} + \sum_k \xi_k^2. \tag{4}$$

Each systematic acts on the prediction through a template, and since the pulls are small by construction we keep only the linear response,

$$\mu_i(\boldsymbol{\theta}, \boldsymbol{\xi}) \simeq \mu_i(\boldsymbol{\theta}) + \sum_k \xi_k v_{ki}, \qquad v_{ki} \equiv \frac{\partial \mu_i}{\partial \xi_k}\bigg|_{\boldsymbol{\xi}=0}, \tag{5}$$

so v_k is simply the shift in the predicted spectrum produced by moving systematic k by one standard deviation. Building the v_k is the whole modelling effort; everything after that is linear algebra.

Profiling Eq. (4) over $\boldsymbol{\xi}$ is then a Gaussian integral, and the result is the familiar statement that pull terms and a covariance matrix are two descriptions of the same thing. Writing $D = \mathrm{diag}(\mu)$ and $\boldsymbol{\Delta} = \boldsymbol{n} - \boldsymbol{\mu}(\boldsymbol{\theta})$, and minimising over $\boldsymbol{\xi}$,

$$\chi^2_{\min}(\boldsymbol{\theta}) = \boldsymbol{\Delta}^T C^{-1} \boldsymbol{\Delta}, \qquad C = D + \sum_k v_k v_k^T = D + V^T V, \tag{6}$$

where V is the matrix whose rows are the v_k. The Poisson variance sits on the diagonal and every systematic adds a rank-one outer product; correlations between bins are carried entirely by the shapes of the v_k. No nuisance is ever minimised numerically.

For a projection we set the data equal to the prediction at the true point, $\boldsymbol{n} = \boldsymbol{\mu}(\boldsymbol{\theta}_0)$, which is what makes it an Asimov dataset. Expanding $\boldsymbol{\mu}(\boldsymbol{\theta})$ to first order about $\boldsymbol{\theta}_0$ turns Eq. (6) into a quadratic form,

$$\Delta\chi^2(\boldsymbol{\theta}) = (\boldsymbol{\theta} - \boldsymbol{\theta}_0)^T F\, (\boldsymbol{\theta} - \boldsymbol{\theta}_0), \qquad F_{ab} = \Big(\frac{\partial \boldsymbol{\mu}}{\partial \theta_a}\Big)^T C^{-1} \Big(\frac{\partial \boldsymbol{\mu}}{\partial \theta_b}\Big), \tag{7}$$

so the projected uncertainty on one parameter with all others profiled is $\sigma(\theta_a) = \sqrt{(F^{-1})_{aa}}$, and for a single parameter alone $\sigma(\theta) = 1/\sqrt{\boldsymbol{d}^T C^{-1} \boldsymbol{d}}$ with $\boldsymbol{d} = \partial \boldsymbol{\mu} / \partial \theta$. This is the expression evaluated for $\sin^2\theta_W$ and for (g_V, g_A) in Section II.

Two practical points make it cheap. First, C is a diagonal matrix plus a low-rank correction, so the Woodbury identity

$$C^{-1} = D^{-1} - D^{-1}V^T\left(\mathbb{1} + VD^{-1}V^T\right)^{-1}VD^{-1} \tag{8}$$

replaces the inversion of an $N_{\text{bins}} \times N_{\text{bins}}$ matrix by that of an $N_{\text{modes}} \times N_{\text{modes}}$ one. With 50 keV binning and a few tens of modes this is the difference between a scan that runs in seconds and one that does not run at all, and it is what allows the sterile analysis of Section III to leave one unconstrained mode free per energy bin. Second, whenever the prediction is affine in a parameter, as it is in the couplings and in $\sin^2 2\theta_{14}$, the templates are built once and every point of a scan is an algebraic combination of them rather than a re-evaluation of the physics.

B. When does the near reactor pile up?

A reactor at tens of metres produces IBD at roughly 1 Hz per 10 MW ($R_{\text{IBD}} = 0.69$ Hz at 10 MW and 50 m, scaling as P/D^2), each event being a prompt hit followed by a delayed neutron capture about 200 μs later. Three distinct pile-up questions arise, and they have very different thresholds.

(i) Accidental IBD-like coincidences. JUNO's measured fiducial singles above 0.7 MeV run at $R_{\text{sing}} = 4.7$ Hz [10] (Section II B), and the reactor adds delayed neutron captures at R_{IBD}. A random pairing of one of each inside the coincidence window τ and the 1.5 m distance cut is an accidental IBD candidate, at a rate

$$R_{\text{acc}} \simeq R_{\text{sing}}\, R_{\text{IBD}}\, \tau\, f_{\text{vol}}, \qquad f_{\text{vol}} \sim \left(\frac{1.5\ \text{m}}{R_{\text{fid}}}\right)^3, \tag{9}$$

Both R_{acc} and the true rate R_{IBD} scale linearly with reactor power, so their ratio,

$$\frac{R_{\text{acc}}}{R_{\text{IBD}}} \simeq R_{\text{sing}}\, \tau\, f_{\text{vol}}, \tag{10}$$

is independent of power and distance altogether and is fixed by JUNO's own singles rate rather than by the source. With $\tau = 1$ ms and $R_{\text{fid}} = 16.5$ m this is 3.5×10^{-6}: 0.21 against 6.0×10^4 events per day at 10 MW and 50 m. Note that this is a different mechanism from the accidentals JUNO already subtracts, which pair two *singles* (0.049 per day in the 59-day release [7]) and do not scale with any reactor at all; the near reactor adds a term that is negligible next to it. Either way this is a background one *measures* from off-window sidebands, not a pile-up limit.

(ii) Genuine IBD self-pile-up. Two reactor IBDs within one coincidence window ($\tau \sim 1$ ms) confuse the prompt-to-delayed pairing. Since the events are Poisson in time, the probability that

a given IBD is accompanied by at least one more within τ is

$$P_{\text{pile}} = 1 - e^{-R_{\text{IBD}}\tau} \simeq R_{\text{IBD}}\,\tau \qquad (R_{\text{IBD}}\tau \ll 1), \tag{11}$$

and this one does *not* cancel: it grows linearly with the source. Numerically it is 7×10^{-4} at 10 MW and 50 m, 0.7% at 100 MW, and 6.7% at 1 GW. This is therefore the operative boundary. Writing $R_{\text{IBD}} \propto P/D^2$ and imposing $P_{\text{pile}} < 1\%$ gives a condition on the source strength alone,

$$\frac{P}{D^2} \lesssim 0.06\,\text{MW}\,\text{m}^{-2}, \qquad \text{i.e.} \qquad P \lesssim 150\,\text{MW}\left(\frac{D}{50\,\text{m}}\right)^2, \tag{12}$$

which corresponds to $\approx 8.6 \times 10^5$ IBD per day whatever the geometry (145 MW at 50 m, 580 MW at 100 m, 2.3 GW at 200 m). Every configuration in this report (10 MW at 50 m, and up to 100 MW in the mobile-reactor study of notebook 2) sits comfortably inside it; a GW-class core would have to stand $\gtrsim 150$ m away.

(iii) Single-hit dead time and DAQ. The reactor's total hit rate (prompt, delayed and EνES) is 1.4 Hz at 10 MW and 50 m, and only 145 Hz even at 1 GW and 50 m, which is 0.01% to 1.5% of the radioactivity singles JUNO already triggers on. Event overlap within a $\sim 1\,\mu$s readout window stays at the $\sim 1\%$ level set by radioactivity, independent of the reactor. Single-hit physics is therefore never rate-limited by the reactor; its reactor-induced backgrounds are the *untagged* IBD positrons and the EνES events themselves, which are modelled explicitly in Sections II to IV. (The reactor's own neutrons and gammas leaking through its shield are a separate, shielding-design question addressed in notebook 8; see Section V.)

The conclusion is that below the boundary of Eq. (12) pile-up is a percent-level nuisance measured in situ, and the physics reach is set by statistics and systematics, not by rate.

II. STANDARD-MODEL PRECISION WITH EνES (NOTEBOOK 5)

A. Cross sections and rates

The observable is elastic scattering off atomic electrons, so it is worth setting up the couplings carefully: the whole measurement is a statement about them. Below the W and Z masses the interaction is a four-fermion contact term, and for $\nu_\alpha e \to \nu_\alpha e$ two diagrams contribute. Every flavour has the neutral current,

$$\mathcal{L}_{\text{NC}} = -\frac{G_F}{\sqrt{2}}\left[\bar{\nu}_\alpha\gamma^\mu(1-\gamma_5)\nu_\alpha\right]\left[\bar{e}\gamma_\mu(g_V - g_A\gamma_5)e\right], \qquad g_V = 2\sin^2\theta_W - \tfrac{1}{2}, \quad g_A = -\tfrac{1}{2}, \tag{13}$$

while for $\alpha = e$ alone there is in addition the charged current, in which the W is exchanged between the ν_e and the electron. A Fierz rearrangement turns that charged-current term into the same $(V - A) \times (V - A)$ structure as Eq. (13) with unit couplings, so the two simply add and the effect of the charged current is the shift

$$C_V = g_V + \delta_{\alpha e}, \qquad C_A = g_A + \delta_{\alpha e}. \tag{14}$$

This is why ν_e and $\bar{\nu}_e$ scatter so differently from $\nu_{\mu,\tau}$, and why a reactor experiment measuring $\bar{\nu}_e e$ is sensitive to a combination of couplings different from the one an accelerator $\nu_\mu e$ experiment measures.

It is convenient to move to chiral couplings, $C_{L,R} = \frac{1}{2}(C_V \pm C_A)$ for neutrinos, because then the two helicity configurations contribute incoherently and the differential cross section takes its standard form. For antineutrinos the roles of L and R interchange, $C_{L,R} = \frac{1}{2}(C_V \mp C_A)$, and

$$\frac{d\sigma}{dT} = \frac{2G_F^2 m_e}{\pi}\left[C_L^2 + C_R^2\left(1 - \frac{T}{E}\right)^2 - C_L C_R \frac{m_e T}{E^2}\right]. \tag{15}$$

The three terms have clean interpretations. The first is helicity-conserving scattering, isotropic in the centre of mass and therefore flat in T. The second is helicity-flipping, suppressed by the $(1 - T/E)^2$ factor that vanishes at the kinematic endpoint. The third is the interference of the two, suppressed by $m_e T/E^2$ and negligible at reactor energies. The kinematic limit follows from two-body kinematics,

$$T_{\max} = \frac{2E^2}{m_e + 2E}, \tag{16}$$

which for $E = 4$ MeV gives $T_{\max} = 3.76$ MeV: the recoil takes almost the whole neutrino energy, so the $T > 1$ MeV analysis window would reach down to $E_\nu = 1.21$ MeV, below the IBD threshold at which the measured flux stops. The reference point used throughout is the PDG 2024 world average of the ν–e couplings, $g_V = -0.040$ and $g_A = -0.507$ [1], which coincides with the radiatively corrected SM prediction (-0.0395, -0.5063 [11]) and corresponds to an effective low-energy $\sin^2\theta_W = (g_V + \frac{1}{2})/2 = 0.230$; sensitivities are quoted as errors on that effective $\sin^2\theta_W$. Cross sections come from Neptune [12], and the parametric (g_V, g_A) implementation used for the scans reproduces its recoil shape to machine precision at Neptune's own tree-level couplings ($\sin^2\theta_W = 0.2223$, $g_A = -\frac{1}{2}$). The absolute normalisation uses $(\hbar c)^2 = 3.8938 \times 10^{-28}\,\mathrm{GeV}^2\,\mathrm{cm}^2$: a shift of this constant is a shift of the EνES rate, and under the IBD anchor of Section II F a 0.7% one would move $\sin^2\theta_W$ by 1.4×10^{-3}, which is the whole error budget. Figure 1 compares the cross sections per carbon of LAB, and Fig. 2 shows the recoil shapes at fixed E_ν. The unit is the

$CH_{1.63}$ of Section I: one carbon nucleus, 1.63 free protons and 7.63 electrons. (I also evaluated the trident channel $\bar{\nu}_e N \to \bar{\nu}_e e^+ e^- N$ with NEPTUNE, coherently on carbon and elastically on the free protons: at reactor energies it is $O(\alpha)$ and phase-space suppressed above its 1.02 MeV threshold and amounts to $\sim 10^{-5}$ of the EνES rate, 0.04 events per day at 10 MW and 50 m, so it is not carried further.) Four features drive the analysis. IBD is $\sim 100\times$ larger per molecule than $\bar{\nu}_e e$ even after the 7.63 electrons are counted, which is why the IBD sample is $20\times$ the EνES one. The antineutrino cross section lies *below* the neutrino one for two reasons that are worth keeping apart. The charged current shifts C_V and C_A by the same $+1$, so it enhances the flat C_L^2 term for ν_e but the phase-space-suppressed $C_R^2(1-T/E)^2$ term for $\bar{\nu}_e$, whose flat coupling $C_L = \frac{1}{2}(C_V - C_A) = \sin^2\theta_W$ is left untouched and is small. On top of that the CC–NC interference is genuinely destructive for $\bar{\nu}_e$: adding the pure-NC and pure-CC rates would give 0.412 in units where the full answer is 0.231. The NC-only $\bar{\nu}_{\mu,\tau}$ channel is a further factor ~ 3 down; it is irrelevant at 50 m but carried anyway. And the $T > 1$ MeV window removes about 60% of the $\bar{\nu}_e e$ rate at $E_\nu = 3$ MeV, less at higher energy. The $\bar{\nu}_e$ recoil spectrum is flat and then falls in T (the $C_R^2(1-T/E)^2$ term), and it is the E_ν-dependence of that fall-off, not the total rate, that carries the $\sin^2\theta_W$ information once the normalisation is free.

The flavour composition is exact at every baseline, here and in Section III: the $\bar{\nu}_{\mu,\tau}$ component from $1 - P_{ee}$ (6×10^{-4} at 50 m, 6% at 1.4 km) is carried with its NC-only cross section. An electron only participates for T above its binding energy (carbon K 284 eV, L 19/11 eV, hydrogen 13.6 eV); this atomic-binding correction [13] is applied to every spectrum and is exactly unity above 1 keV, so it is completely negligible for the analysis window $T \in [1, 6.5]$ MeV.

The resulting rates for 10 MW at 50 m are 5.95×10^4 IBD per day ($\sim 1700\times$ the far reactors) and 3.2×10^3 EνES per day in the analysis window, 99.99% of them $\bar{\nu}_e$. Because the flux stops at the IBD threshold and $T_{\rm max}(1.806\,\mathrm{MeV}) = 1.58$ MeV, 48% of that EνES rate sits in a part of the window the flux model cannot fill; the quoted rate is therefore a lower bound there, and the model carries an artificial edge at 1.58 MeV. It costs nothing: closing the window at $T > 1.6$ MeV instead of $T > 1$ MeV moves $\sigma(\sin^2\theta_W)$ from 0.137 to 0.138, because what the joint fit measures is a rate ratio in which both channels share the same flux. Figure 3 shows the recoil spectrum with the backgrounds I model: IBD singles (untagged-neutron fraction 1%, 577 per day, measurable in situ from the 99%-tagged sample), solar neutrinos (^{8}B, hep and CNO continua plus the pep line [14, 15], LMA flavour-weighted, 164 per calendar day), and the detector's own singles, which dominate everything else and are the subject of Section II B.

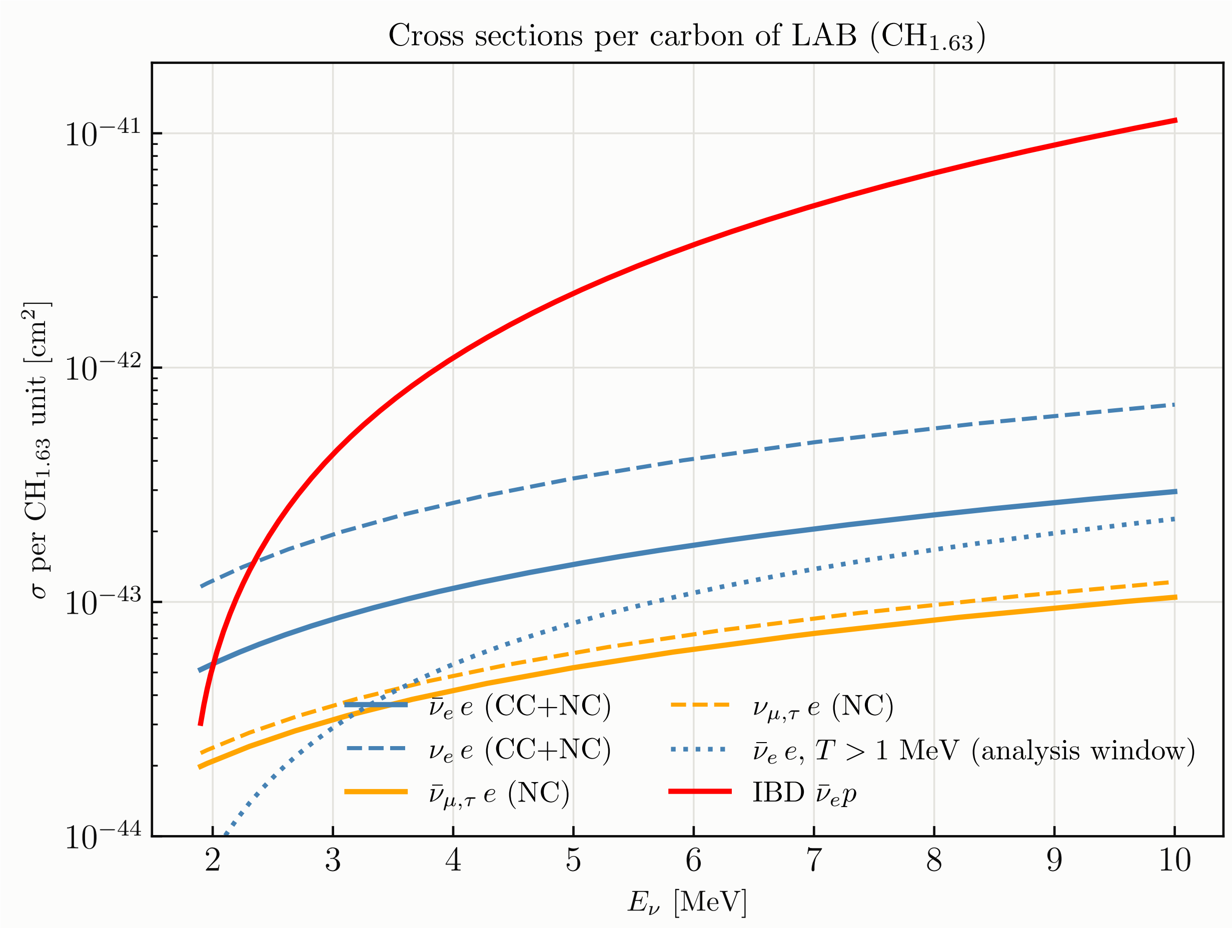

FIG. 1: Cross sections per carbon of LAB (one $CH_{1.63}$ unit: 1 carbon, 1.63 free protons, 7.63 electrons): EνES for $\bar{\nu}_e/\nu_e$ (CC+NC) and $\bar{\nu}_{\mu,\tau}/\nu_{\mu,\tau}$ (NC), the $\bar{\nu}_e e$ rate restricted to the $T > 1$ MeV analysis window, and IBD on the free protons [4].

B. Detector singles, and the reactor-off measurement

The dominant single-hit background is not the reactor but JUNO itself, and two recent measurements pin it down. The final radioactivity assessment [10] reports, from first data, a fiducial singles rate of 4.7 Hz above 0.7 MeV for $R < 17.2$ m, better than the 7.2 Hz design budget. The delivered scintillator has U/Th $\lesssim 3.4 \times 10^{-17}$ g/g, some 30× purer than the 10^{-15} requirement, so the internal component collapses to $\lesssim 0.1$ Hz and $\sim 95\%$ *of the fiducial singles are external* γ from the acrylic vessel, the PMT glass and the steel truss. On the solar side [16] the same backgrounds set JUNO's own ^{8}B threshold at 2 MeV, through cosmogenic ^{11}C.

The spectral consequence is what makes the measurement possible (Fig. 4). An external γ can deposit at most its own line energy, so the external component *stops at the* 2.615 MeV ^{208}Tl

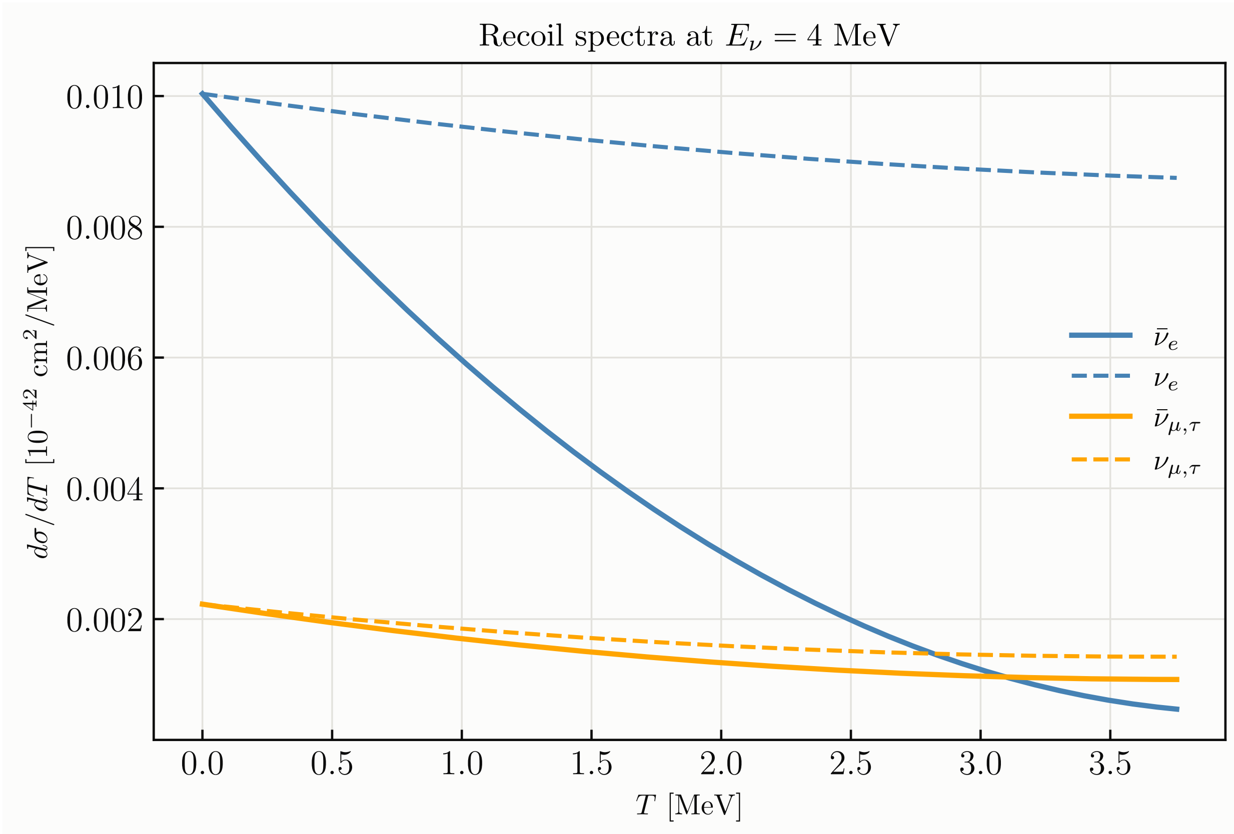

FIG. 2: Recoil spectra $d\sigma/dT$ at $E_\nu = 4$ MeV for the four flavour/helicity channels; the antineutrino–neutrino difference is the CC–NC interference.

line. An internal decay contains its whole cascade, so internal ^{208}Tl reaches 5.0 MeV, but only at ~ 84 decays per day in 20 kt at the measured purity. Across the 2.6 MeV wall the singles fall by three orders of magnitude, from 3.6×10^4 per day in 2–3 MeV to 26 per day in 3–4 MeV, and the signal-to-background turns over from 0.02 to 9.

I model this in `reactor/singles.py`: external γ from a composite line list with Compton continua, internal decays from the *measured* concentrations (so ^{214}Bi runs at ~ 730 and ^{208}Tl at ~ 84 per day in 20 kt), and the cosmogenic isotopes ^{11}C, ^{10}C, ^{11}Be, ^{12}B, ^{6}He, ^{8}Li and ^{9}Li with a JUNO-like veto suite. The *rate* is anchored to the measurement, but the *shape* is a component model rather than a transport simulation, the radial profile is an effective exponential tuned to their Fig. 6 rather than a γ attenuation length (their own reconstruction systematic on the FV singles is 20–30%), and the cosmogenic rates are depth-scaled estimates good to a factor of a few. These numbers should not be taken at face value, so the analysis is built not to depend on them.

a. Turning the reactor off. A parked research reactor can be switched off; the singles cannot. A reactor-off run therefore measures the entire non-reactor background in situ, bin by bin, with

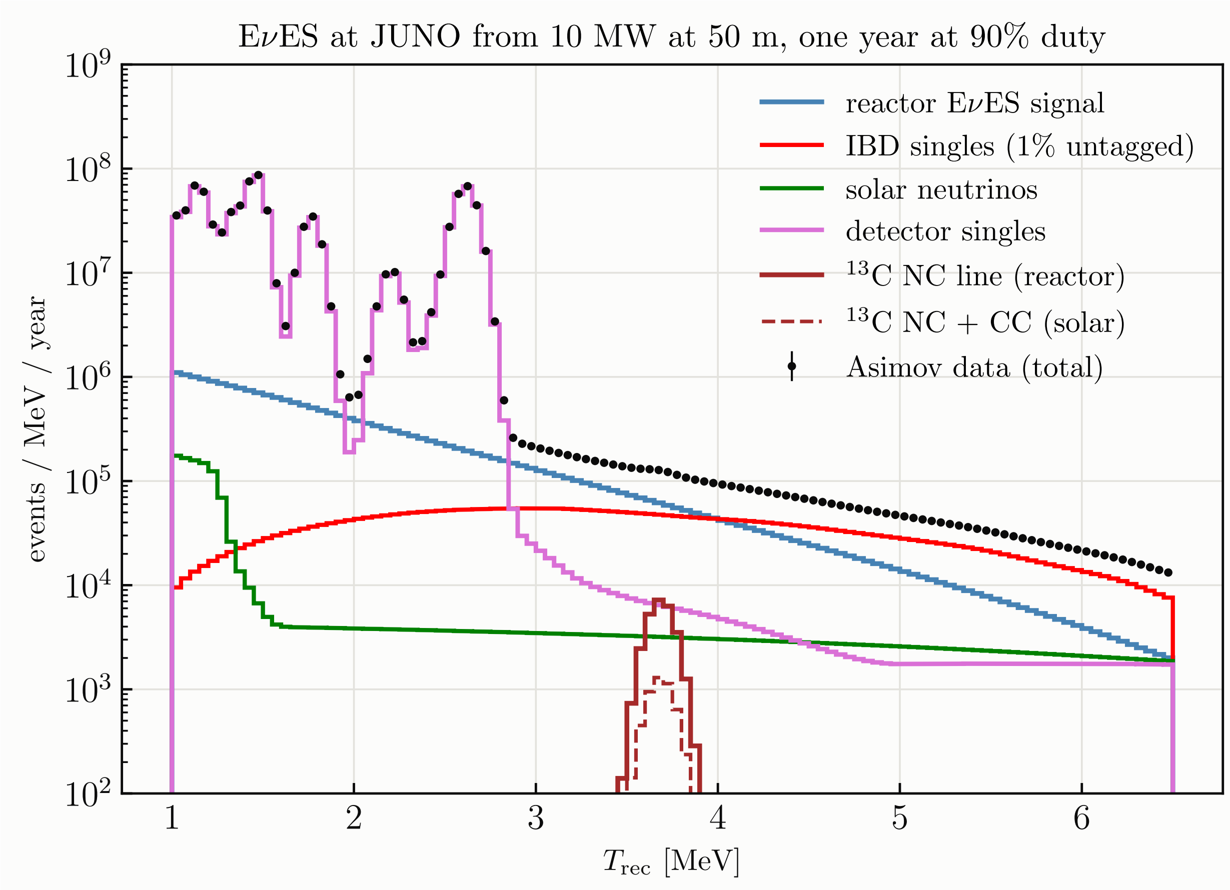

FIG. 3: Asimov EνES recoil spectrum with the modelled backgrounds, per calendar year at 90% duty, for 10 MW at 50 m ($\approx 1.1 \times 10^6$ signal, 1.9×10^5 IBD singles, 6×10^4 solar, 4.5×10^7 detector-singles and 1.4×10^3 ^{13}C NC events per year in the window).

no model at all, and it is worth working through what that is worth, because the answer is what protects everything above from the provisional rates.

Take one bin and split the reactor-on prediction into three pieces: the signal S, the reactor-correlated background R (the untagged IBD positrons), and the non-reactor background B (singles, solar, and the solar ^{13}C of Section II C). Running for a time t with the reactor on and rt with it off gives two measurements,

$$\langle N^{\mathrm{on}} \rangle = (S + R + B)\, t, \qquad \langle N^{\mathrm{off}} \rangle = B\, rt, \tag{17}$$

and we now treat B as a completely free parameter, one per bin, with no prior whatsoever. That is the crucial modelling choice. We are not assuming the singles level, and we are not assuming their spectral shape either, since a free normalisation in every bin subsumes any distortion one could write down.

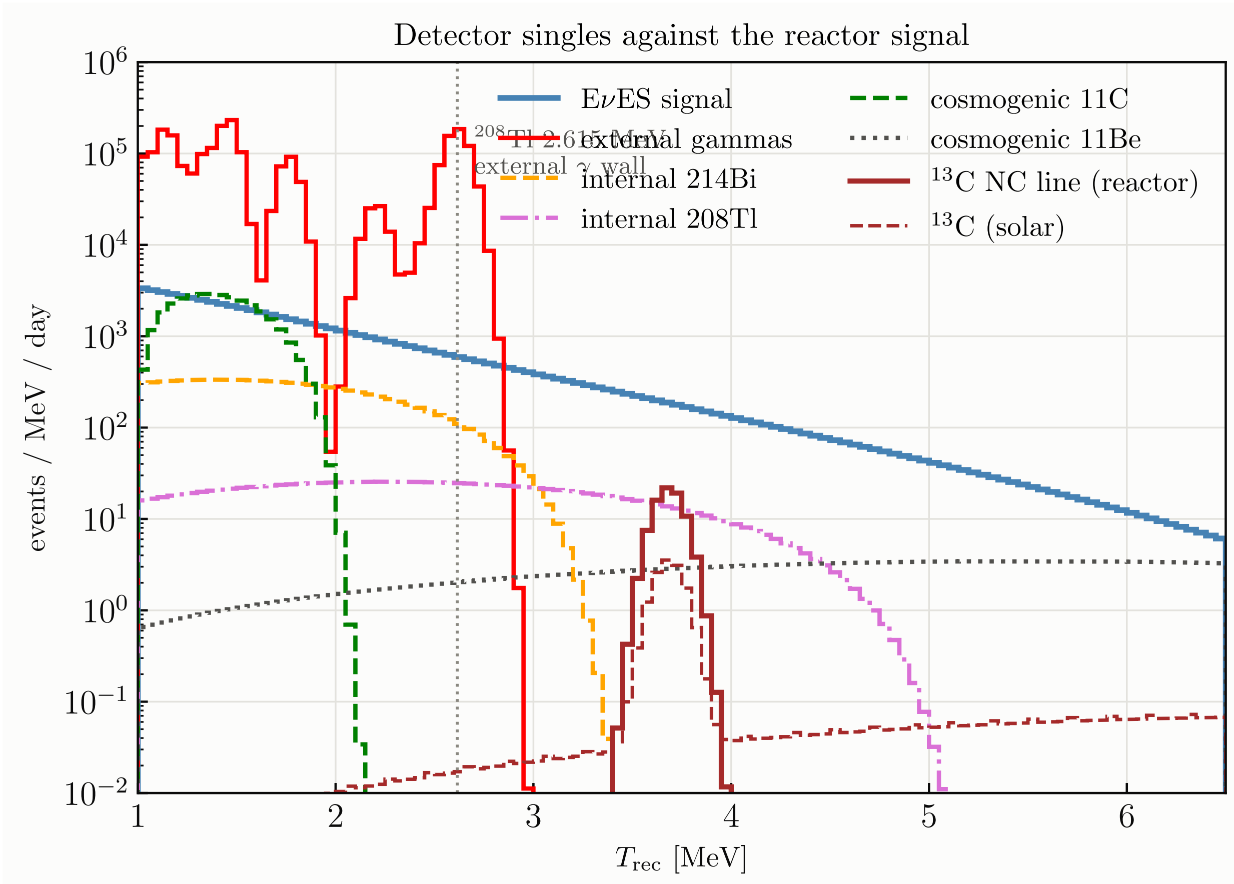

FIG. 4: Detector singles by component against the EνES signal at $R = 16.5$ m. The external γ component ends at the ^{208}Tl line; above it only internal ^{208}Tl (full 5.0 MeV cascade), the cosmogenics and the ^{13}C lines survive. The reactor-driven ^{13}C NC line at 3.685 MeV clears the wall by a megaelectronvolt.

Let δ be a shift of B away from its nominal value. It moves the on-prediction by δt and the off-prediction by $\delta r t$, so with x and y the residuals of the two measurements and $V_{\rm on} = (S{+}R{+}B)t$, $V_{\rm off} = Brt$ their variances,

$$\chi^2(\delta) = \frac{(x - \delta t)^2}{V_{\rm on}} + \frac{(y - \delta r t)^2}{V_{\rm off}}. \tag{18}$$

Setting $\partial\chi^2/\partial\delta = 0$ gives

$$\hat{\delta}\, t = \frac{x/V_{\rm on} + r\, y/V_{\rm off}}{1/V_{\rm on} + r^2/V_{\rm off}}, \tag{19}$$

and substituting back collapses the two terms into one,

$$\chi^2_{\rm min} = \frac{\left(x - y/r\right)^2}{V_{\rm on} + V_{\rm off}/r^2}. \tag{20}$$

The numerator is exactly the subtracted estimator one would have written down by hand, the off-sample scaled by $1/r$ to match the on-exposure, and the denominator is its variance. Writing that variance out,

$$V_{\rm on} + \frac{V_{\rm off}}{r^2} = (S+R+B)\,t + \frac{B\,rt}{r^2} = \Big[S+R+B\Big(1+\frac{1}{r}\Big)\Big]t. \tag{21}$$

So profiling a free per-bin background is not merely tractable, it is a one-line change to the analysis: every background normalisation systematic disappears and the Poisson term is inflated by B/r. That is how it is implemented, as an inflation of the diagonal D in Eq. (6) rather than as $N_{\rm bins}$ extra modes, which keeps C numerically well conditioned even when B exceeds S by two orders of magnitude.

Three things follow directly from Eq. (21). Equal on and off exposure, $r = 1$, gives the familiar factor of two, $\mathrm{Var} = S+R+2B$. The result depends on the singles only through $\sqrt{B}$, never through their assumed level or shape, which is exactly the protection needed given how provisional those rates are. And the limit $r \to 0$ diverges, as it must: with no off-sample and a free background in every bin, the signal is perfectly degenerate with B and there is no measurement at all. A short off-run is therefore worse than none: it inflates the variance by B/r while buying almost no sample, and a 2%-length off-run costs 45% on the joint fit. “No off-run” is a different analysis rather than the $r \to 0$ limit of this one — there the singles *shape* comes from the model of Section II B and only its normalisation is profiled, which gives 0.135 against 0.137 at $r = 1$ — and any number quoted that way is a statement about how far one trusts a background simulation, not a statement about data.

The price of $r = 1$ is calendar time, since half the wall clock is spent with the reactor off and a given MW yr takes twice as long to deliver. That is also what makes $r = 1$ the right choice rather than merely an adequate one: at fixed *calendar* time the live exposure is $\propto 1/(1+r)$, so minimising $(1+r)\big[S+R+B(1+1/r)\big]$ gives

$$r_{\rm opt} = \sqrt{\frac{B}{S+R+B}} \;\longrightarrow\; 1 \quad (B \gg S), \tag{22}$$

and $B \simeq 1.2 \times 10^5$ per day against $S \simeq 3.2 \times 10^3$ puts this measurement firmly in that limit. Figure 5 shows what the off-run buys, and Table I that it works.

Two more things follow. The answer is remarkably insensitive to the singles level (Table I): 10× the background costs 16% and 30× costs 41%. And the fiducial radius barely matters: $\sigma(\sin^2\theta_W)$ moves only from 0.134 to 0.139 between $R = 15.5$ and 17.0 m while the singles rate inside the cut moves by a factor of six, because the reactor-off subtraction, not the cut, is doing the work.

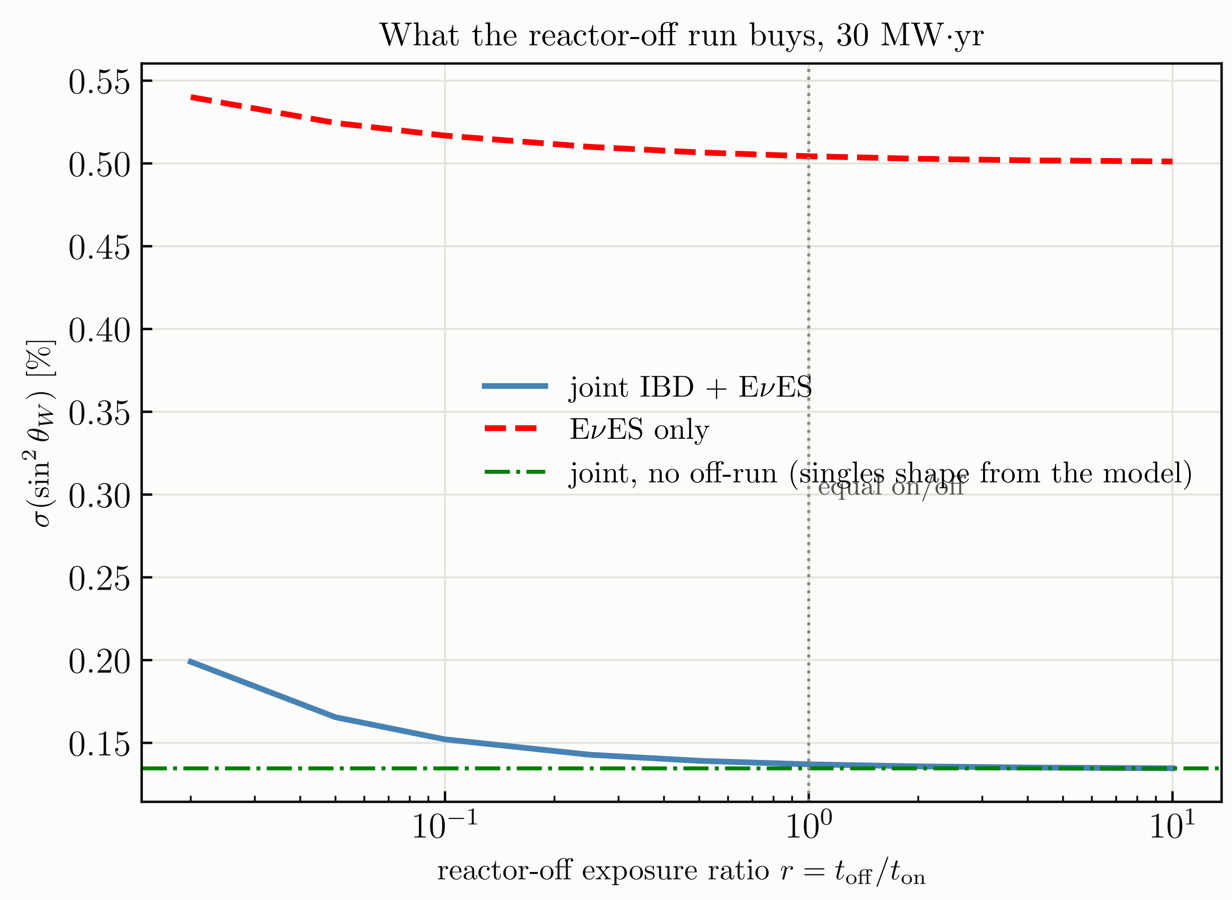

FIG. 5: $\sigma(\sin^2\theta_W)$ against the reactor-off exposure ratio at 30 MW yr. Equal on/off running ($r = 1$) is within 2% of the $r \to \infty$ asymptote and is the optimum at fixed calendar time, Eq. (22); a 2%-length off-run costs 45% on the joint fit. The horizontal line is the separate analysis in which there is no off-run at all and the singles shape is taken from the model.

I therefore keep the $R = 16.5$ m of the rest of this report rather than optimising a cut against provisional rates.

C. Neutrino interactions on ^{13}C

Natural carbon is 1.1% ^{13}C by atom, so the scintillator holds ~ 210 t of it, 9.7×10^{30} nuclei, the target JUNO uses for its model-independent ^{8}B measurement [16]. Both channels land in the single-hit window, and they behave very differently here.

a. The neutral current is purely isovector axial. At the quark level the neutral current is

$$\mathcal{L}_{\rm NC} = -\frac{G_F}{\sqrt{2}} \left[\bar{\nu}\gamma^\mu(1-\gamma_5)\nu\right] \sum_q \bar{q}\gamma_\mu(g_V^q - g_A^q\gamma_5)q, \tag{23}$$

singles rate assumed	rate [day^{-1}]	EνES-only	joint
none	0	0.474	0.120
0.1× nominal	1.2×10^4	0.492	0.130
nominal (as modelled)	1.2×10^5	0.504	0.137
3× nominal	3.7×10^5	0.510	0.144
10× nominal	1.2×10^6	0.520	0.159
30× nominal	3.7×10^6	0.535	0.193

TABLE I: Robustness of $10^2\,\sigma(\sin^2\theta_W)$ at 30 MW yr to the assumed singles level, with equal reactor-on and reactor-off exposure. A background 30× larger than modelled costs 41%; this is the payoff of measuring it rather than predicting it.

with $g_V^u = \frac{1}{2} - \frac{4}{3}s_W^2$, $g_A^u = +\frac{1}{2}$, $g_V^d = -\frac{1}{2} + \frac{2}{3}s_W^2$ and $g_A^d = -\frac{1}{2}$, $s_W^2 \equiv \sin^2\theta_W$. Rearranged into isospin components this is the familiar statement

$$J_\mu^Z = J_\mu^3 - 2s_W^2\,J_\mu^{\rm em}, \qquad J_\mu^3 = V_\mu^3 - A_\mu^3, \tag{24}$$

and because the electromagnetic current is purely vector, *the entire axial part of the neutral current is $-A_\mu^3$, the third isospin component of the axial current, carrying coefficient exactly unity and no s_W^2 whatsoever*. Only the vector part is mixed and s_W^2-suppressed. At the nucleon level, in the non-relativistic limit, the space components of A_μ^3 reduce to

$$A_k^3 \;\longrightarrow\; \frac{g_A}{2}\sum_i \sigma_{k,i}\,\tau_i^3 \;+\; \frac{\Delta s}{2}\sum_i \sigma_{k,i}, \tag{25}$$

an isovector spin operator with coefficient $g_A = 1.2754$ and an isoscalar one carried by the strange-quark spin, $\Delta s \approx -0.08$, an order of magnitude smaller.

Now the transition does the rest of the work. The ^{13}C ground state is $J^\pi = 1/2^-$ and the excited level is $3/2^-$: $\Delta J = 1$ with no parity change. The leading (allowed) operator of the *vector* current is the Fermi charge operator $\sum_i \tau_i^3$, which is $\Delta J = 0$ and therefore cannot connect the two states at all; the vector current enters only through weak magnetism, $\propto \mu_V\,\boldsymbol{\sigma}\times\boldsymbol{q}/2M_N$, suppressed by $q/2M_N \sim 10^{-3}$ at these energies. The leading operator of the *axial* current is precisely the $\Delta J = 1$ spin operator of Eq. (25). So the line strength is

$$\sigma_{\rm NC}(E_\nu) = \frac{G_F^2}{\pi}\,(E_\nu - E_x)^2\,\frac{g_A^2}{4}\,B({\rm GT}^0)\Big[1 + \mathcal{O}\big(q/M_N,\ \Delta s/g_A\big)\Big], \qquad B({\rm GT}^0) \equiv \frac{\big|\langle f\|\sum_i \sigma_i\tau_i^3\|i\rangle\big|^2}{2J_i+1}, \tag{26}$$

with $E_x = 3.685$ MeV and the $(E_\nu - E_x)^2$ from the phase space of the outgoing massless neutrino. The 1/4 is the $\tau^3/2$ normalisation of the isospin current.

Two consequences are worth stating plainly. First, *the line carries no leading dependence on* $\sin^2\theta_W$ *at all*: it measures $g_A^2 B(\mathrm{GT}^0)$, the isovector axial strength. It is therefore not a second weak-mixing-angle measurement but a genuinely different observable, sensitive to the hadronic axial current where EνES is sensitive to the electron couplings, and to any new physics that touches quark neutral currents without touching electrons. Second, the same isovector spin operator governs the $M1$ strength of the 3.685 MeV level, so $B(\mathrm{GT}^0)$ is constrained by nuclear data rather than being a free parameter.

b. *Why the isoscalar axial current is so much harder.* It is worth spelling out what the neutral current does *not* contain. In Eq. (23) the axial couplings satisfy $g_A^u = -g_A^d = +\frac{1}{2}$, so the up and down isoscalar axial contributions cancel identically and, within u, d, s,

$$A_\mu^Z = \tfrac{1}{2}\big(\bar{u}\gamma_\mu\gamma_5 u - \bar{d}\gamma_\mu\gamma_5 d\big) - \tfrac{1}{2}\,\bar{s}\gamma_\mu\gamma_5 s \;\longrightarrow\; \tfrac{1}{2} g_A\,\tau^3 \;-\; \tfrac{1}{2}\Delta s, \tag{27}$$

so the entire isoscalar axial piece is carried by the strange-quark spin. That makes it a clean probe of $\Delta s \approx -0.08$, and hence of the proton spin decomposition, but it also makes it small: $\Delta s/g_A \approx 1/16$.

There are only two ways to get at it, and neither survives reactor energies. One can isolate it with a transition that the isovector operator cannot drive at all, namely $T = 0 \to T = 0$ with $\Delta J = 1$ in a self-conjugate nucleus, since τ^3 is an isovector and cannot connect two $T = 0$ states. The textbook case is the 12.71 MeV $1^+, T = 0$ level of ^{12}C, measured against the isovector 15.11 MeV $1^+, T = 1$ level. But the rate then costs $(\Delta s/g_A)^2 \approx 4\times10^{-3}$, and 12.71 MeV sits above the reactor endpoint of ~ 10 MeV in any case; the low-lying $T = 0 \to T = 0$ alternatives (^{6}Li at 4.31 MeV, ^{10}B at 3.59 MeV) are either particle-unbound or present in far too small a quantity. The alternative is to let the isoscalar amplitude *interfere* with the isovector one, which buys linear rather than quadratic sensitivity but requires both to reach the same final state, i.e. a free nucleon. That is elastic $\nu p \to \nu p$, the BNL E734 measurement. At reactor energies the proton recoil is $T_p = 2E_\nu^2/M_p \lesssim 0.14$ MeV, quenched to ~ 10 keV of visible energy in scintillator, far below any threshold. The isoscalar axial current is therefore out of reach here, and the ^{13}C line remains an isovector measurement.

c. *The charged current is neutrino-only, and empirically normalised.* For $\nu_e + {}^{13}\mathrm{C} \to e^- + {}^{13}\mathrm{N(g.s.)}$ the allowed cross section is

$$\sigma_{\mathrm{CC}}(E_\nu) = \frac{G_F^2\cos^2\theta_C}{\pi}\, p_e E_e\, F(Z{=}7, E_e)\big[B(\mathrm{F}) + g_A^2 B(\mathrm{GT})\big], \qquad E_e = E_\nu - Q + m_e, \tag{28}$$

with E_e the total electron energy and $Q = 2.22$ MeV the threshold, which is just the ^{13}N–^{13}C atomic mass difference, the electron created and the one shed by the change in Z cancelling. Here

^{13}N(g.s.) is the isobaric analog of ^{13}C(g.s.) (a $T = 1/2$ doublet with $T_3 = \pm 1/2$), so the Fermi matrix element is unquenched, $B(\mathrm{F}) = 1$, and the combination $B(\mathrm{F}) + g_A^2 B(\mathrm{GT})$ is fixed directly by the measured ft value of the mirror $^{13}\mathrm{N} \to {}^{13}\mathrm{C}\, e^+ \nu_e$ decay. The CC cross section is thus known without a nuclear model.

Crucially this channel needs a *neutrino*. The antineutrino partner is $\bar{\nu}_e + {}^{13}\mathrm{C} \to e^+ + {}^{13}\mathrm{B}$, and its threshold follows from the nuclear masses: the atomic mass excesses give $\Delta M_{\mathrm{at}} = 16.562 - 3.125 = 13.437$ MeV, converting to nuclear masses adds one electron mass since Z drops from 6 to 5, and creating the positron costs another, so

$$E_{\mathrm{th}} = \left[M_{\mathrm{nuc}}(^{13}\mathrm{B}) - M_{\mathrm{nuc}}(^{13}\mathrm{C})\right] + m_e = \Delta M_{\mathrm{at}} + 2m_e = 14.46 \text{ MeV}. \tag{29}$$

That is half again the reactor endpoint of ~ 10 MeV, so the reactor contributes nothing to it and the CC channel is a purely solar background, tagged besides by the 863 s ^{13}N β^+ coincidence. The neutral current, by contrast, is flavour blind and identical for neutrinos and antineutrinos, so *the reactor antineutrinos excite it.* The level lies below the 4.946 MeV neutron separation energy, so it is particle bound and de-excites by a single 3.685 MeV γ: a monoenergetic line, 73 keV wide at JUNO's resolution, landing a full MeV above the 2.615 MeV wall in the cleanest part of the window.

d. Rates. I take the energy dependence from Eqs. (26) and (28) and fix the two overall scales by JUNO's own ^{8}B yields, 3032 NC and 3929 CC events in ten years, rather than by evaluating $B(\mathrm{GT}^0)$ from a shell model. Carrying that law from the ^{8}B weighting (3.7–15 MeV) down to the reactor's (3.7–9 MeV) is a modest extrapolation, so the reactor rate should be read as good to a factor of order unity. It gives 4.1 events per day at 10 MW and 50 m, five times the solar NC rate, and as a background to EνES it is negligible: including all of the ^{13}C moves $\sigma(\sin^2\theta_W)$ by one digit in the fifth place. Note that the ^{13}C count cancels exactly between the anchoring and the prediction, so the rate does not depend on it.

As a measurement in its own right it is more interesting. Treating the line normalisation as the parameter of interest, with its cross-section prior dropped and every other systematic standing, the reactor-driven line reaches $\mathbf{12}\boldsymbol{\sigma}$ at 30 MW yr (4.5×10^3 events; 7.0σ at 10 MW yr, 21σ at 100 MW yr). It is the one place in the window where a sharp feature sits on a smooth continuum, so it is limited by statistics rather than by any shape systematic.

D. Neutrino interactions on deuterium

Natural hydrogen carries 156 ppm of deuterium, so JUNO's 1.44×10^{33} free protons come with 2.2×10^{29} deuterons, about 750 kg of D, 2.3% of the ^{13}C count. Deuterium is the target one would most like to have: the neutral-current breakup $\nu + d \to \nu + n + p$ is a pure Gamow-Teller transition on the simplest of all nuclei, with a matrix element known from pionless effective field theory to about 1% rather than from a shell model, and it is how SNO measured the total solar flux. Two channels are open, and their thresholds are worth assembling piece by piece because the gap between them is what decides everything that follows. Both must first pay the deuteron binding energy,

$$B_d = m_p + m_n - M_d = 938.272 + 939.565 - 1875.613 = 2.224 \text{ MeV}. \tag{30}$$

The neutral current pays nothing else, since the outgoing neutrino is massless and the final state is just an unbound np pair. The charged current must in addition turn a proton into a neutron and materialise a positron,

$$\bar{\nu}_e + d \to \bar{\nu}_e + n + p, \quad E_{\text{th}} = B_d = 2.224 \text{ MeV}, \tag{31}$$

$$\bar{\nu}_e + d \to e^+ + n + n, \quad E_{\text{th}} = B_d + (m_n - m_p) + m_e = 2.224 + 1.293 + 0.511 = 4.028 \text{ MeV}, \tag{32}$$

the second of which is just $2m_n + m_e - M_d$ written in a more transparent way. Folding these against the reactor spectrum, the neutral current samples 71.4% of the flux and the charged current only 14.5% of it, a factor of five that the cross sections more than undo: the charged current is the larger of the two per neutrino, by a factor ~ 2.5 at these energies, as SNO's two channels also show.

As for ^{13}C the energy dependence is derived and the two overall scales are external. Both channels are allowed transitions, so a lepton of energy E_ℓ recoils against a two-body continuum of relative energy ϵ whose density of states goes as $\sqrt{\epsilon}$,

$$\sigma_{\text{NC}} \propto \int_0^Q (Q - \epsilon)^2 \sqrt{\epsilon}\, d\epsilon \propto Q^{7/2}, \qquad \sigma_{\text{CC}} \propto \int_0^{Q'} p_e E_e \sqrt{\epsilon}\, d\epsilon, \tag{33}$$

with $Q = E_\nu - B_d$ and $E_e = Q' - \epsilon + m_e$, $Q' = E_\nu - 4.028\,\text{MeV}$. This omits the $^1S_0$ final-state interaction, which enhances small ϵ in *both* channels and so largely cancels between them. The scales are the per-fission cross sections at ^{235}U-dominated reactors, well matched to the HALEU core, $\sigma_{\text{CC}} = 1.06 \times 10^{-44}$ and $\sigma_{\text{NC}} = 5.6 \times 10^{-45}\,\text{cm}^2/\text{fission}$, on which theory and the reactor measurements agree to about 10%; every number below inherits that 10% and nothing worse. The raw rates at 10 MW and 50 m are then 0.157 (CC) and 0.084 (NC) events per day.

The two channels then part company completely, and it is the signature rather than the rate that decides. The charged current gives a prompt positron followed by *two* neutron captures, a triple coincidence, and the two neutrons share a vertex. The only serious accidental is an ordinary IBD event picking up a second neutron from another IBD, and requiring the second capture within 1 m and 1 ms of the first reduces that to 9×10^{-3} per day against a signal of 0.157 per day. That is 172 events per 30 MW yr at 13σ, and the significance is signal-statistics limited rather than background limited. Note that both signal and accidental scale with reactor power, so the signal-to-background is a property of the vertex cut and not of the exposure. The background that would need real work is cosmogenic fast neutrons, which genuinely produce multi-neutron events.

The neutral current, by contrast, produces a lone 2.22 MeV neutron-capture γ with no prompt at all, since the proton recoil is sub-MeV and heavily quenched. Against 5×10^3 singles per day in the surrounding 0.45 MeV, and even with equal reactor-off subtraction, this reaches 0.02σ. It is inaccessible. The channel one wants is precisely the one the abundance denies: a deuterated target would be needed, and doping the scintillator to 1% D would give 10 CC and 5 NC events per day, while a dedicated one-tonne D_2O cell at 10 m would give 1.1 and 0.6. Neither is a modification to JUNO; both are separate experiments. What the parked reactor does deliver for free is a respectable $\bar{\nu}_e d$ charged-current measurement riding on the same data as everything else.

E. Systematics

The EνES-only fit carries the following modes: a 2% normalisation (power$\times$target$\times$efficiency); the EνES cross-section normalisation at 0.2% (the $O(\alpha)$ radiative corrections, the only theory error left once the couplings are the parameters of interest); the 25-mode measured ^{235}U shape covariance [3]; ^{238}U at $\pm$15%; the IBD-singles normalisation at $\pm$10%; solar at $\pm$3%; energy scale, bias and resolution at 0.5/0.5/5%, coherent between the electron and positron responses, plus a 0.2% scale mode on the electron response alone; and the fuel-evolution (Pu-ingrowth) mode at $\pm$30%. The detector singles carry no normalisation prior at all: under Eq. (21) their level and shape are measured, and what remains is a 1% coherent *stability* residual between the on and off periods (radon drift, muon-rate variation). That residual is a complete null: even a 10% residual leaves $\sigma(\sin^2\theta_W)$ unchanged at five digits, because the steeply-falling singles shape is nearly orthogonal to the coupling-induced distortion of the recoil spectrum.

The joint fit stacks the IBD anchor spectrum (1.0–9.0 MeV in 50 keV bins) with the EνES spectrum, *frees the flux normalisation* (a loose 10% prior) and lets the IBD channel measure it.

The transfer to EνES is then limited by four terms: the channel ratio $N_e/N_p \times \varepsilon$ (0.5%), the IBD cross-section normalisation (0.2%, from τ_n and radiative corrections [4, 17]) with a zero-mean linear tilt on top (0.1%), the EνES cross-section normalisation (0.2%), and the residual e^-/e^+ energy-scale difference (0.2%). The last of these deserves a word. Freeing the flux normalisation is what lets the anchor pin the energy response, and it does so only to the extent that the electron and positron scales move together; they are calibrated with different sources, since an electron deposits one continuous track and a positron its kinetic energy plus two 511 keV quanta. The decorrelated residual is therefore carried, and it costs 8% of the final number. Under the anchor the standard JUNO backgrounds (far reactors, geoneutrinos, ^{9}Li/^{8}He, world reactors, ^{214}Bi–Po and "other", about 45 per day against 5.8×10^4 per day) enter at their release priors [7], and the shared ^{235}U shape modes act on both channels coherently, so the anchor constrains them in situ.

F. Results

Table II decomposes $\sigma(\sin^2\theta_W)$ at 30 MW yr, and Fig. 6 shows the exposure dependence.

Three structural conclusions come out of this. First, *the EνES-only measurement is limited by the flux shape*: the recoil shape breaks the normalisation degeneracy far better than the naive rate argument ($|d \ln R/d \sin^2\theta_W| = 4.75$) suggests, but the 2–4% correlated bin uncertainties of the measured ^{235}U spectrum imitate the coupling-induced shape change and saturate the precision at $\sigma(\sin^2\theta_W) \approx 5 \times 10^{-3}$ regardless of exposure. Second, *the joint fit removes that wall*: with the normalisation freed and measured by the anchor, the ^{235}U shape covariance and the coherent energy pulls become near-nulls (the 7×10^7-event anchor pins both), and what remains is the transfer budget, then the detector singles, then the fuel-evolution mode and the solar/JUNO backgrounds. The result, $\sigma(\sin^2\theta_W) = 1.4 \times 10^{-3}$ at 30 MW yr and 8.0×10^{-4} at the plausible best of the transfer inputs, is a $\sim$ 0.6%-of-itself low-energy weak mixing angle, competitive with the best existing determinations [11].

Third, and this is worth stating plainly because it says what the measurement actually requires: *the joint number is a rate ratio and almost nothing else.* Adding the four transfer priors in quadrature and dividing by $|d \ln R/d \sin^2\theta_W|$ gives $\sqrt{0.5^2 + 0.2^2 + 0.2^2 + 0.2^2}\%/4.75 = 1.3 \times 10^{-3}$, which is 92% of the full answer; the spectral information contributes the rest. The exposure dependence of Fig. 6 flattens for the same reason. So the claim of this section is conditional in a specific and testable way: *if* $N_e/N_p \times \varepsilon$ can be established to 0.5% and $\sigma_{\rm IBD}$ to 0.2%, the parked reactor delivers $\sigma(\sin^2\theta_W) = 1.4 \times 10^{-3}$. Establishing the first of those is a scintillator-assay and efficiency

EνES-only variant, 30 MW yr	$10^2\,\sigma(\sin^2\theta_W)$
all systematics	0.504
no flux-shape covariance	0.352
no detector singles	0.474
no solar/JUNO backgrounds	0.503
no energy scale/bias/res. pulls	0.490
no fuel-evolution mode	0.502
signal statistics only	0.011
with the measured backgrounds	0.030
joint IBD+EνES, defaults	0.137
IBD xsec 0.5% (Strumia–Vissani)	0.163
channel ratio 0.2%	0.103
e^-/e^+ relative scale 0.1%	0.130
all four transfer terms at best	0.080
no ^{235}U shape covariance	0.133
no detector singles	0.120
2%-length reactor-off run ($r = 0.02$)	0.199
no reactor-off run, singles shape modelled	0.135

TABLE II: Precision on $\sin^2\theta_W$ for 30 MW yr near-reactor exposure (≈ 3.3 calendar years with about 90% uptime). Entries are 10^2 times the *absolute* error, so 0.137 means $\sigma(\sin^2\theta_W) = 1.4\times10^{-3}$, or 0.6% of $\sin^2\theta_W$ itself. "All four transfer terms at best" is IBD xsec 0.1%, channel ratio 0.2%, EνES xsec 0.1% and relative scale 0.1%.

question, not a neutrino one, and it is where the effort belongs.

Figure 8 shows the same anchoring in the (g_V, g_A) plane: $\sigma(g_V) = 0.0038$ and $\sigma(g_A) = 0.0072$ at 30 MW yr, with principal semi-axes 0.0026×0.0077. The singles cost g_A far more than g_V, as they must: g_A is fixed by the low-recoil shape, which is exactly where they live.

Figure 7 places this in the existing landscape. Electron-flavour experiments see the CC amplitude too, so their cross sections depend on $(g_V+1,\, g_A+1)$ and constrain *bands* along the resulting degeneracies rather than ellipses: $\nu_e e$ (LSND [18], LAMPF [19]) bounds $(g_V+g_A+2)^2+(g_V-g_A)^2/3 = 2.34\pm0.35$, and $\bar\nu_e e$ (TEXONO [20]) bounds $(g_V-g_A)^2+(g_V+g_A+2)^2/3$ through $\xi = \sigma/\sigma_{\mathrm{SM}} = 1.08\pm0.26$. Muon-flavour scattering (CHARM II [21]: $g_V = -0.035\pm0.017$, $g_A = -0.503\pm0.017$; BNL E734 [22]: -0.107 ± 0.045, -0.514 ± 0.036) is pure NC and gives ellipses; the PDG world average, $g_V = -0.040\pm0.015$ and $g_A = -0.507\pm0.014$ [1], is domi-

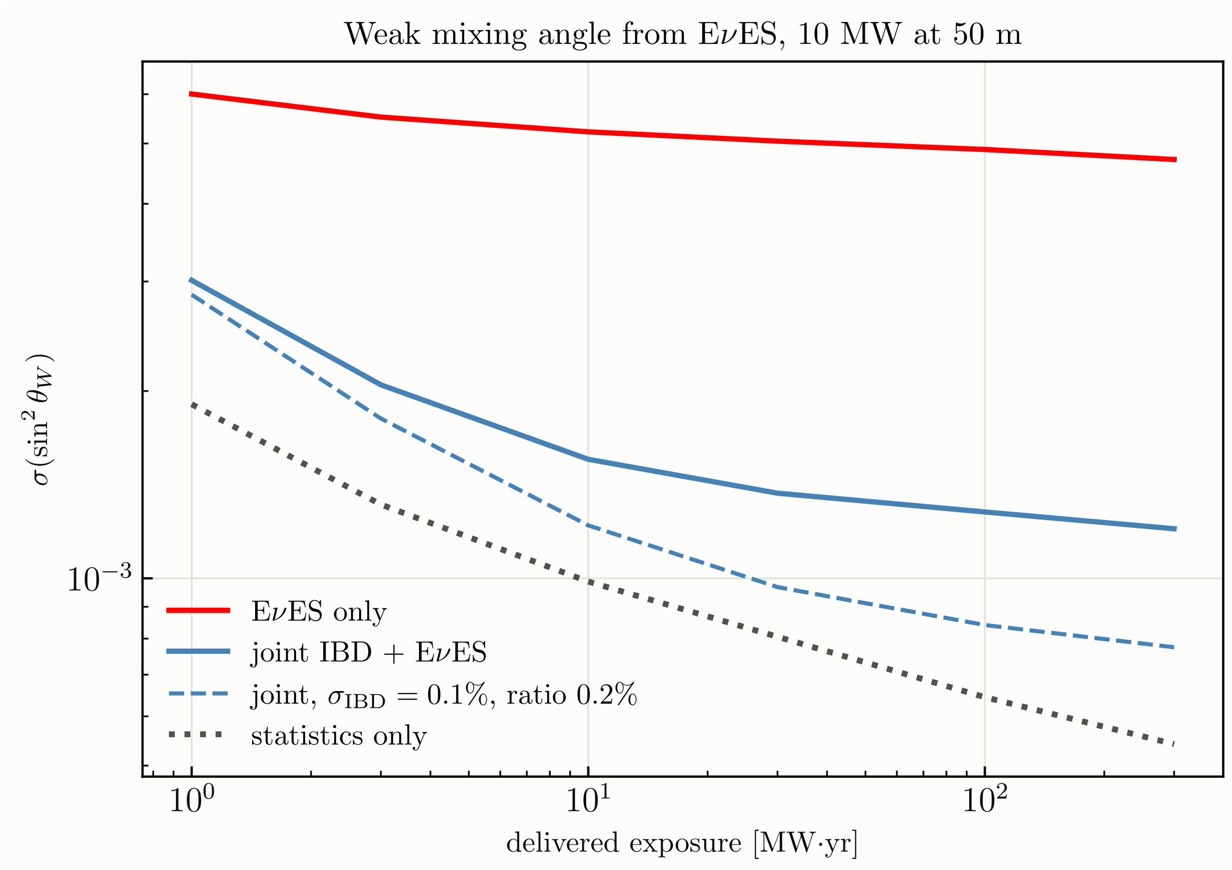

FIG. 6: $\sigma(\sin^2\theta_W)$ versus delivered exposure. The EνES-only fit saturates on the measured flux-shape covariance; the IBD-anchored fit converts statistics into precision until it reaches its own floor, the transfer budget.

nated by CHARM II and assumes flavour universality. Neutrino trident production (CCFR [23], $\sigma/\sigma_{\rm SM} = 0.82 \pm 0.28$; CHARM II [24], 1.58 ± 0.57) constrains the ν_μ–*muon* couplings [25] rather than ν–e, so I do not draw it. Reactor $\bar{\nu}_e e$ data combined give $\sin^2\theta_W = 0.252 \pm 0.030$ [26]. Our measurement is $\bar{\nu}_e$: it lands on TEXONO's band and shrinks it by nearly two orders of magnitude, and its joint 30 MW yr ellipse sits inside the CHARM II and world-average circles, 4.5× tighter in g_V and 2.4× in g_A (and 2.2–6.7× along its principal axes), from a different flavour. All contours are centred on the PDG world-average point.

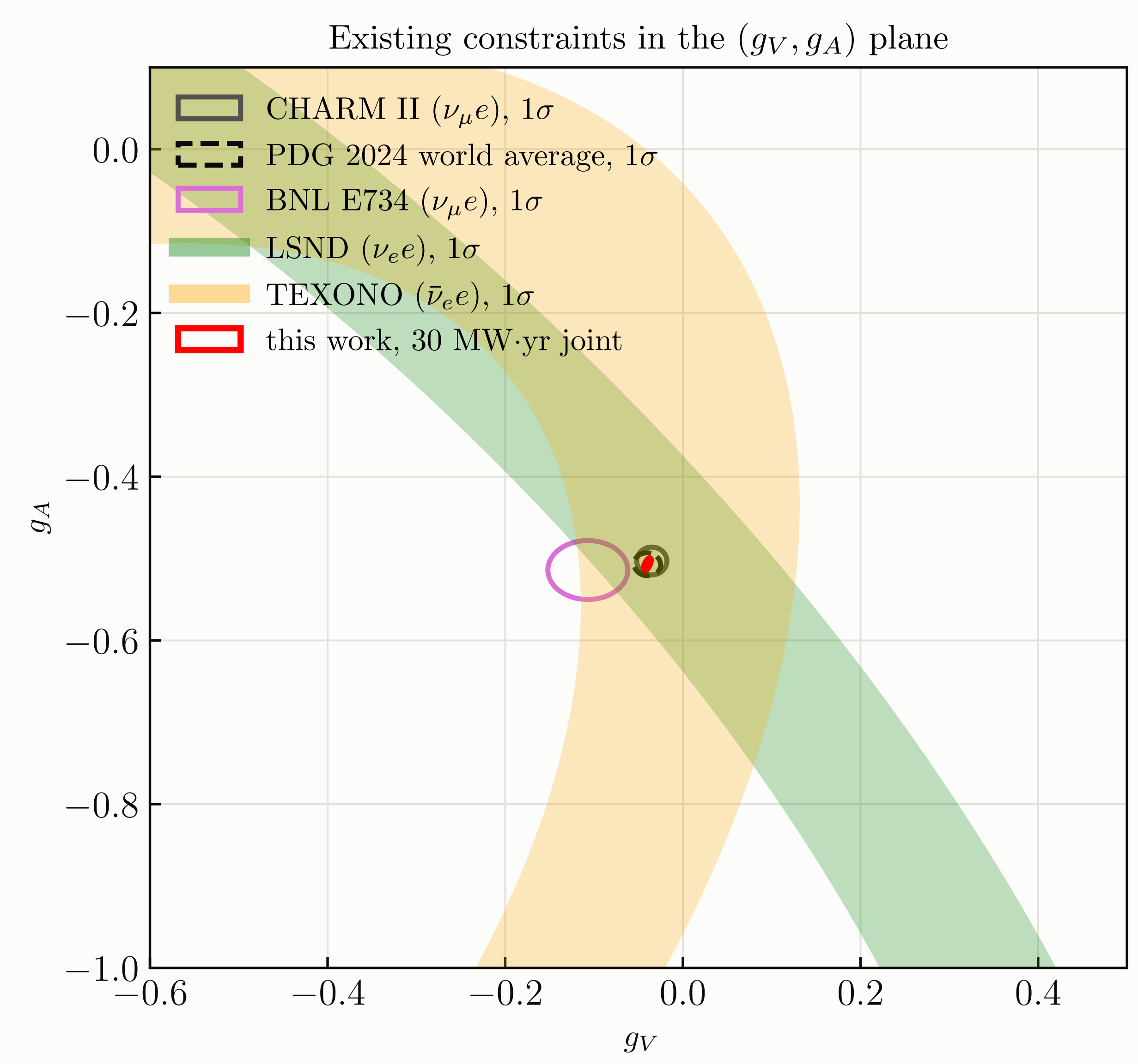

FIG. 7: Existing 1σ constraints in the (g_V, g_A) plane: the $\nu_e e$ (LSND) and $\bar{\nu}_e e$ (TEXONO) bands, the $\nu_\mu e$ ellipses (CHARM II, BNL E734) and the PDG world average, with this work's 30 MW yr joint region at the PDG point.

III. STERILE-NEUTRINO WIGGLES ACROSS THE FINITE DETECTOR (NOTEBOOK 6)

A. The idea

Start from the geometry, because it is the whole reason this works. Put a point source at distance D from the centre of a fiducial sphere of radius R and ask how much target sits at baseline L. The locus of points at fixed L from the source is a sphere, and the part of it inside the detector is

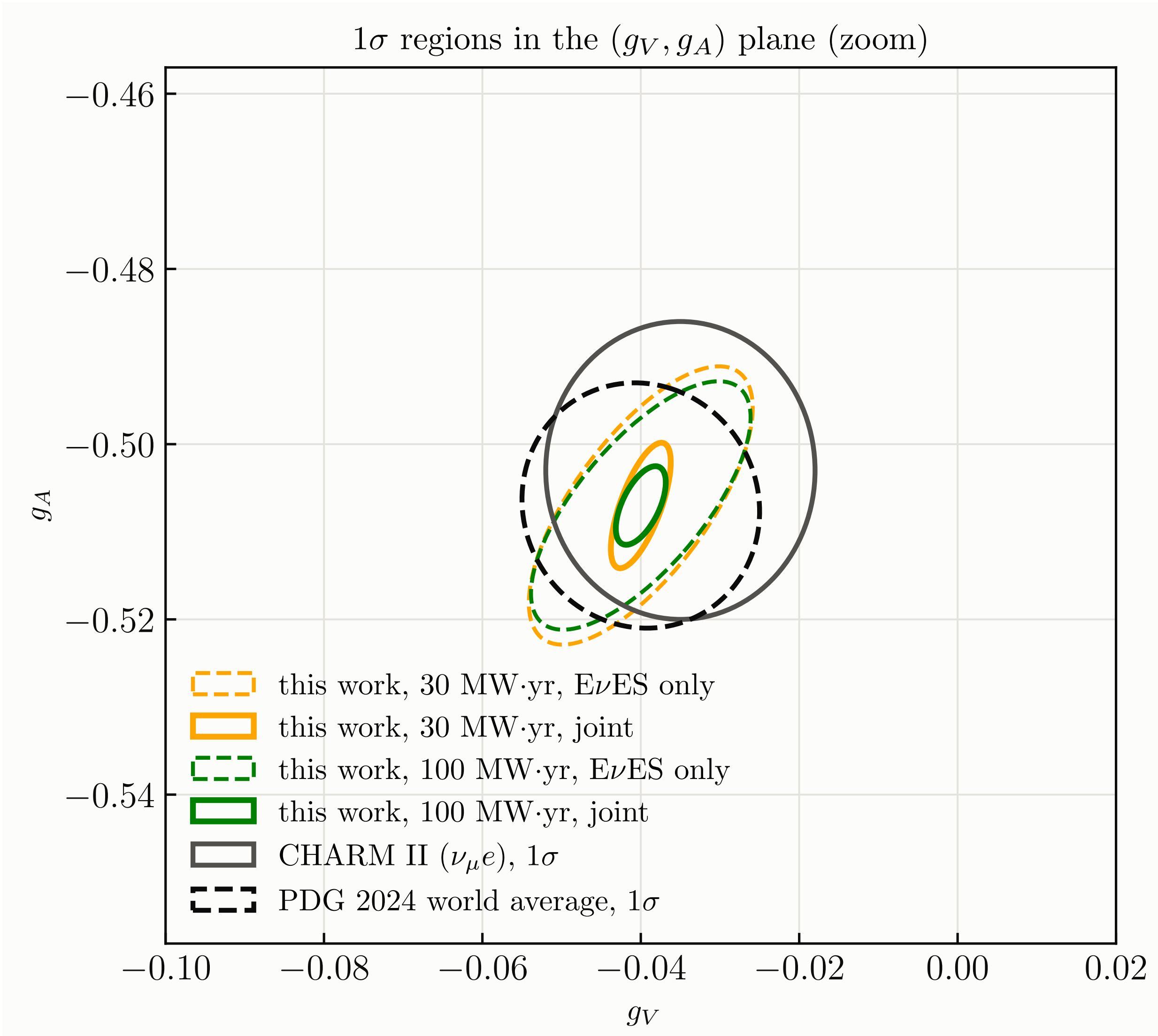

FIG. 8: Zoom on the PDG point: 1σ Fisher regions of this work, EνES-only (dashed) and IBD-anchored (solid), against the CHARM II and PDG world-average ellipses.

a spherical cap. With α the polar angle measured from the source-to-centre axis, the cap edge is where the two spheres intersect, given by the cosine rule as

$$\cos\alpha_{\max}(L) = \frac{L^2 + D^2 - R^2}{2LD}, \tag{34}$$

so the cap area is $2\pi L^2 \left[1 - \cos\alpha_{\max}\right]$ and the target volume between L and $L + dL$ is

$$A(L) = 2\pi L^2 \left[1 - \frac{L^2 + D^2 - R^2}{2LD}\right], \qquad |D - R| \le L \le D + R. \tag{35}$$

The event density then carries the usual inverse-square dilution,

$$\frac{dN}{dL} \propto \frac{A(L)}{L^2} = 2\pi \left[1 - \frac{L^2 + D^2 - R^2}{2LD}\right], \tag{36}$$

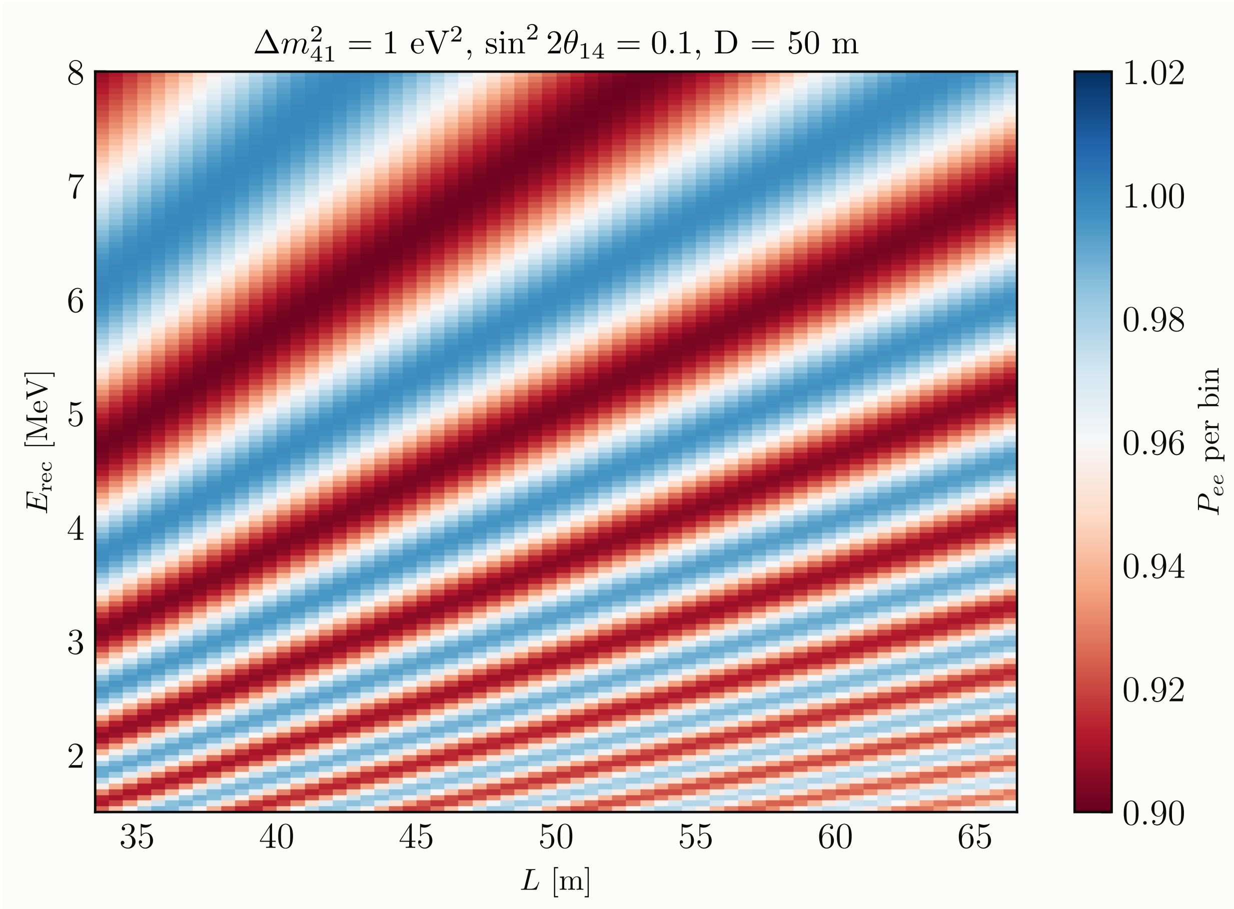

FIG. 9: Survival probability per $(L, E_{\rm rec})$ bin for $\Delta m^2_{41} = 1\,\mathrm{eV}^2$, $\sin^2 2\theta_{14} = 0.1$, $D = 50$ m.

which is exact, contains no free parameter, and is worth pausing on: the L^2 of the cap area cancels the $1/L^2$ of the flux, leaving a distribution fixed entirely by D and R. Nothing about the reactor enters it. At $D = 50$ m and $R = 16.5$ m the baseline spans 33 to 67 m, a factor of two. For $\Delta m^2_{41} \sim 0.1$–$10\,\mathrm{eV}^2$ the oscillation length at reactor energies is metres to tens of metres, so the disappearance pattern *wiggles across the detector* (Figs. 9 and 10). Binning events in reconstructed vertex baseline L makes JUNO a many-baseline experiment in one fill, and no flux or cross-section systematic can follow: the geometry is exact and the source spectrum carries no L-dependence.

B. Oscillation, smearing, statistics

I use the full 3+1 vacuum survival probability,

$$P_{ee} = 1 - (1 - s_{14})^2\,(1 - P^{3\nu}_{ee}) - 4s_{14}(1 - s_{14}) \sum_{i=1}^{3} |U_{ei}|^2_{3\nu} \sin^2 \Delta_{i4}, \qquad s_{14} = \sin^2\theta_{14}, \tag{37}$$

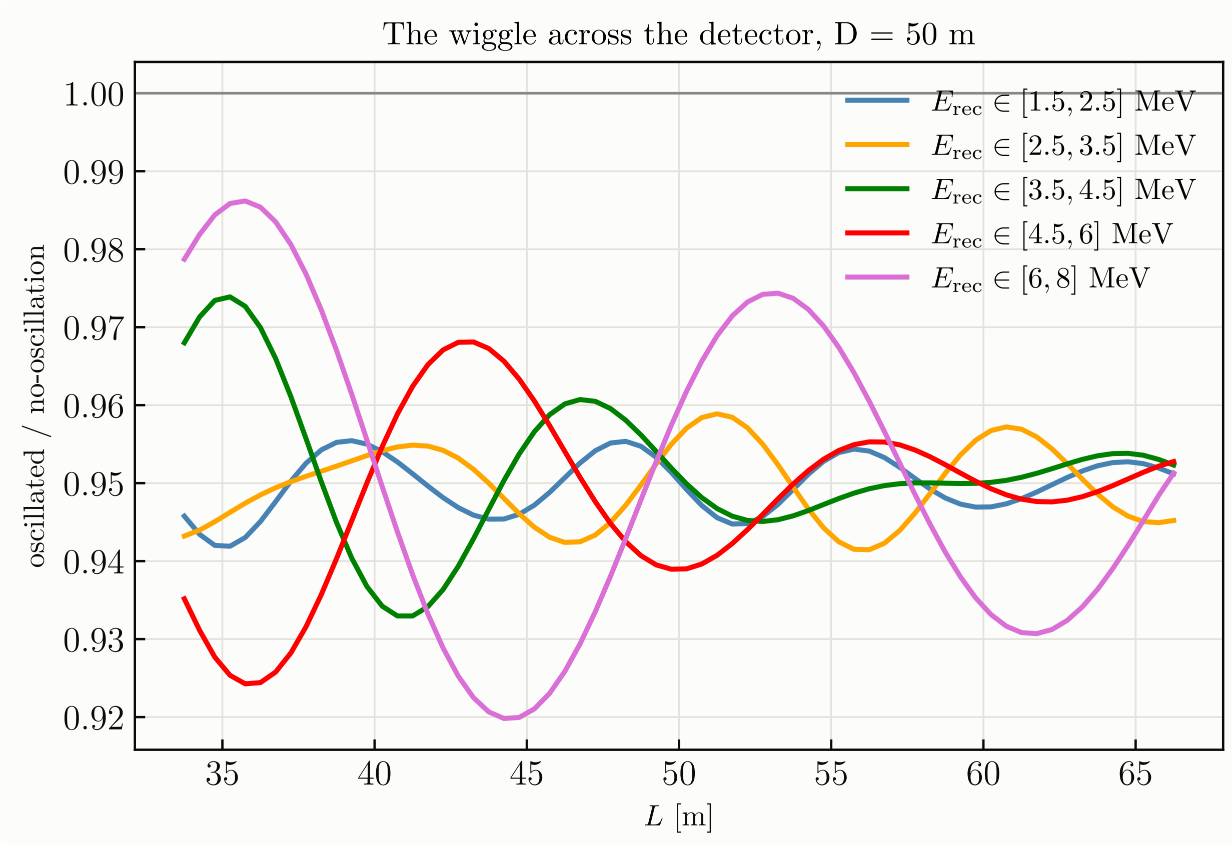

FIG. 10: The same benchmark: oscillated/unoscillated ratio versus L in five energy windows across the spectrum. The wavelength scales as E (~ 5 m at 2 MeV, ~ 20 m at 7 MeV), so different energies sample the same L range with different phase advance; this (L, E) pattern is what no source systematic can mimic.

with the exact three-flavour $P_{ee}^{3\nu}$ in the Δm^2_{ee} parameterisation [27] and $\Delta_{i4} = 1.267\,(m_4^2 - m_i^2)L/E$ ($\Delta m^2_{31} = \Delta m^2_{ee} + s^2_{12}\Delta m^2_{21}$). I validated it against a brute-force PMNS-amplitude computation to 8.5×10^{-6}. Expanding the $(1 - s_{14})^2$ prefactor, $1 - (1 - s_{14})^2 = \frac{1}{2}\sin^2 2\theta_{14} + O(\sin^2 2\theta_{14}^2)$, so the prediction is exactly affine in $\sin^2 2\theta_{14}$ to first order with the template

$$W = \sum_i |U_{ei}|^2_{3\nu} \sin^2 \Delta_{i4} - \tfrac{1}{2}\big(1 - P_{ee}^{3\nu}\big), \tag{38}$$

the second term being the piece of the ordinary θ_{13} disappearance that the sterile mixing switches off. It is 6×10^{-4} of the first at 50 m and $\sim 3\%$ at 1.4 km, and it carries no L-wiggle. At 50 m the whole expression reduces to $1 - \sin^2 2\theta_{14} \sin^2 \Delta_{41}$, while at 1.4 km, where $\Delta_{31} \approx 1$ rad, the three sterile phases separate and the null hypothesis is the θ_{13}-oscillated spectrum. IBD (CC) and EνES (CC+NC) deplete identically, since an oscillated $\bar{\nu}_e$ is sterile; the $\bar{\nu}_{\mu,\tau}$ that the standard oscillation produces still scatters by NC and is carried with its own cross section, so the EνES null is exact at every standoff.

Two effects smear the true baseline at fixed reconstructed vertex, and both damp the wiggle in the same way, so it is worth deriving the damping once. Write the oscillating factor as $\sin^2 kL$ with $k = 1.267\,\Delta m^2/E$ in units of eV^2, MeV and metres, and suppose the true L is Gaussian-distributed about the reconstructed value with variance σ_L^2. Using $\sin^2 kL = \frac{1}{2}[1 - \cos 2kL]$ and the standard Gaussian average of a cosine, $\langle \cos 2kL \rangle = \cos(2k\bar{L})\, e^{-2k^2\sigma_L^2}$,

$$\left\langle \sin^2 kL \right\rangle = \tfrac{1}{2}\Big[1 - \cos(2k\bar{L})\, e^{-2k^2\sigma_L^2}\Big]. \tag{39}$$

The mean survives untouched and only the oscillating part is attenuated, by $e^{-2k^2\sigma_L^2}$. Counts are therefore conserved by the smearing, which is why the null spectra are unaffected by it and only the signal template is damped.

The two contributions to σ_L^2 add in quadrature. A finite core of radius a, uniformly emitting, has a line-of-sight coordinate distributed as the projection of a uniform ball, whose variance is $\langle z^2 \rangle = \frac{1}{V}\int z^2\, d^3r = a^2/5$. The vertex resolution contributes σ_1^2/E for a JUNO-like $\sigma_{\mathrm{vtx}} = \sigma_1/\sqrt{E}$. Hence

$$\sigma_L^2 = \frac{\sigma_1^2}{E} + \frac{a^2}{5}, \tag{40}$$

and since the exponent goes as $k^2 \propto (\Delta m^2/E)^2$, the damping switches on quadratically in Δm^2. This is what terminates the reach at high Δm^2, and Eq. (40) shows immediately which term does it: the core radius enters with no energy dependence, so once $a^2/5$ exceeds σ_1^2/E the source, not the detector, sets the wall. Averaging over the finite width of an L bin is done exactly with sub-nodes rather than analytically.

The statistics then simplify remarkably. The sterile term in P_{ee} enters multiplied by $\sin^2 2\theta_{14}$ and nothing else, so the predicted spectrum is affine in that one parameter,

$$\boldsymbol{\mu}(\sin^2 2\theta_{14}) = \boldsymbol{\mu}_0 - \sin^2 2\theta_{14}\, \boldsymbol{W}(\Delta m^2), \tag{41}$$

where $\boldsymbol{\mu}_0$ is the three-flavour null and $\boldsymbol{W}$ is a single wiggle template per Δm^2, built once. Putting Eq. (41) into the Asimov quadratic form Eq. (7) with $\boldsymbol{\theta}_0$ the null,

$$\Delta\chi^2(\Delta m^2, \sin^2 2\theta_{14}) = \sin^2 2\theta_{14}^2\, \boldsymbol{W}^T C^{-1} \boldsymbol{W} \;\equiv\; \sin^2 2\theta_{14}^2\, Q(\Delta m^2), \tag{42}$$

so a single number Q per Δm^2 carries the entire sensitivity and the exclusion curve is closed-form,

$$\sin^2 2\theta_{14}^{\mathrm{excl}}(\Delta m^2) = \sqrt{\frac{\Delta\chi^2_{\mathrm{crit}}}{Q(\Delta m^2)}}, \qquad \Delta\chi^2_{\mathrm{crit}} = 5.99\ (95\%\ \mathrm{CL},\ 2\ \mathrm{dof}). \tag{43}$$

No scan over $\sin^2 2\theta_{14}$ is needed at all, which is what makes the four-standoff, many-mode analysis of Fig. 11 tractable.

C. Why the analysis is shape-only, and what that costs

The Daya Bay ^{235}U spectrum was *measured* at ~ 500 m: a sterile with $\Delta m^2_{41} \sim 0.1$–$3\,\mathrm{eV}^2$ would have depleted it by $\approx \sin^2 2\theta_{14}/2$ in normalisation (4.8–5.0% at $\sin^2 2\theta_{14} = 0.1$, computed) and left a $\lesssim 1\%$ energy tilt. Using its normalisation or shape as a prior would inject the very signal being searched for. The standard analysis therefore leaves the flux normalisation *completely free* and the reactor E-shape *completely free* (one unconstrained mode per E and per T bin, coherent across L; 263 modes in total). Only the L-direction wiggle structure at fixed E can then carry sterile information, and it carries essentially all of it: relative to a Daya-Bay-prior analysis the limits move by less than 0.5% across 0.03–$3\,\mathrm{eV}^2$, and by 11% only at the $10\,\mathrm{eV}^2$ wall. The Daya Bay spectrum serves only as a shape template for event rates. The remaining nuisances are the detector response non-uniformities across the volume, the fuel evolution ($\pm 30\%$, a null), and the IBD/EνES ratio (0.5%). The non-uniformities are the systematics that live in the wiggle direction and both are modelled as Gaussian-correlated fields over L with a 1.5 m correlation length, so that finer binning cannot manufacture high-frequency freedom. One is an *efficiency* field (0.5%), which is energy independent and is therefore broken by the E-dependence of the oscillation phase. The other is a position-dependent *energy scale* (0.2%) — the light collection against radius — which does distort the spectrum in E as a function of L, the same joint direction the signal lives in, and is the more dangerous of the two. It is still not the limiter: the field spans only ~ 20 correlated modes across a 33 m span, which the fit projects out, so its cost saturates and 0.2% and 1% give the same limit to three digits. Together the two fields cost under 1% of the reach. The backgrounds of Section II (IBD singles, solar, and the standard JUNO components distributed as volume in L) are included and shift the limits by less than 0.1%: being smooth in both L and E, they cannot make wiggles.

D. Results

Figure 11 shows the 95% CL reach for 27 MW yr at each of five standoffs against the existing landscape (DANSS [28], PROSPECT [29], STEREO [30], RENO+NEOS [31], Gallium 2σ [32, 33], KATRIN 2025 [34], and the IsoDAR@Yemilab projection [35]). Figure 12 isolates what the finite size buys, and Fig. 13 the high-Δm^2 wall.

The findings, in order of importance:

- The vertex-binned search reaches $\sin^2 2\theta_{14} = 9.5 \times 10^{-4}$ at $\Delta m^2 = 0.5\,\mathrm{eV}^2$, stays below 1.1×10^{-3} over 0.26–$1.1\,\mathrm{eV}^2$ and below 2×10^{-3} from 0.13 to $2.7\,\mathrm{eV}^2$: an order of magnitude

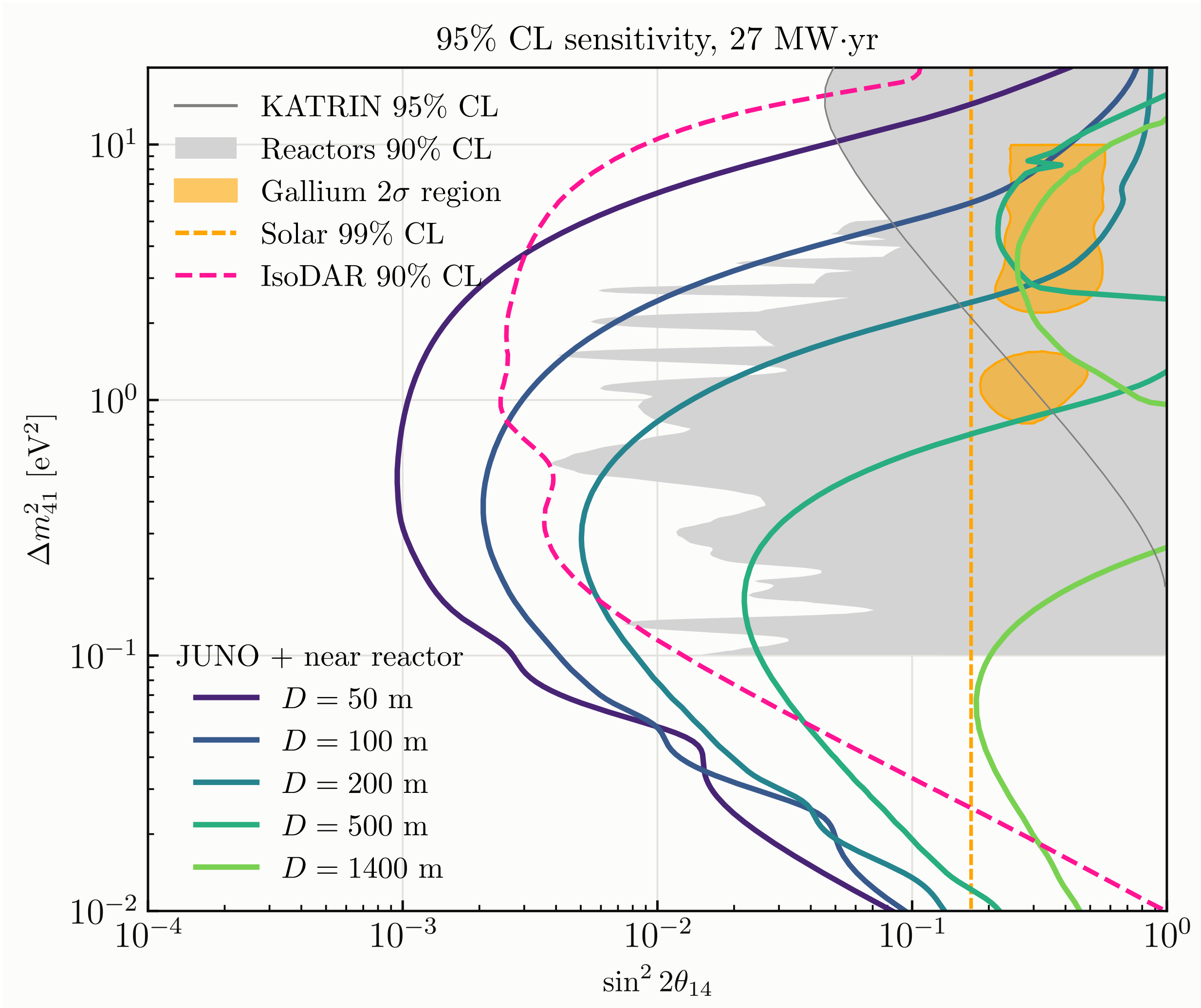

FIG. 11: 95% CL sensitivity in the $\bar{\nu}_e$-disappearance plane for 27 MW yr at $D = 50$, 100, 200, 500 and 1400 m (each standoff separately), against the reactor short-baseline exclusions (DANSS, PROSPECT, STEREO, RENO+NEOS; grey), KATRIN 2025, the Gallium 2σ region, the solar bound, and the IsoDAR@Yemilab projection.

beyond the reactor short-baseline experiments, and covering the Gallium 2σ islands (which span 0.8–10 eV^2) in full.

- *An integrated analysis of the same data, with the same free flux shape, has no sensitivity at all.* One free mode per E bin and a single L bin leaves nothing to fit, and its curve sits at $\sin^2 2\theta_{14} \approx 0.5$–$0.8$ over the whole range where the L-binned search has reach, rising further at the edges (Fig. 12). The only integrated analysis with any reach is one that imposes the Daya Bay flux shape — the measurement that would itself have carried the depletion — and the vertex-binned search beats *that* by ~ 3–4 at its sweet spot near 1–3 eV^2, where

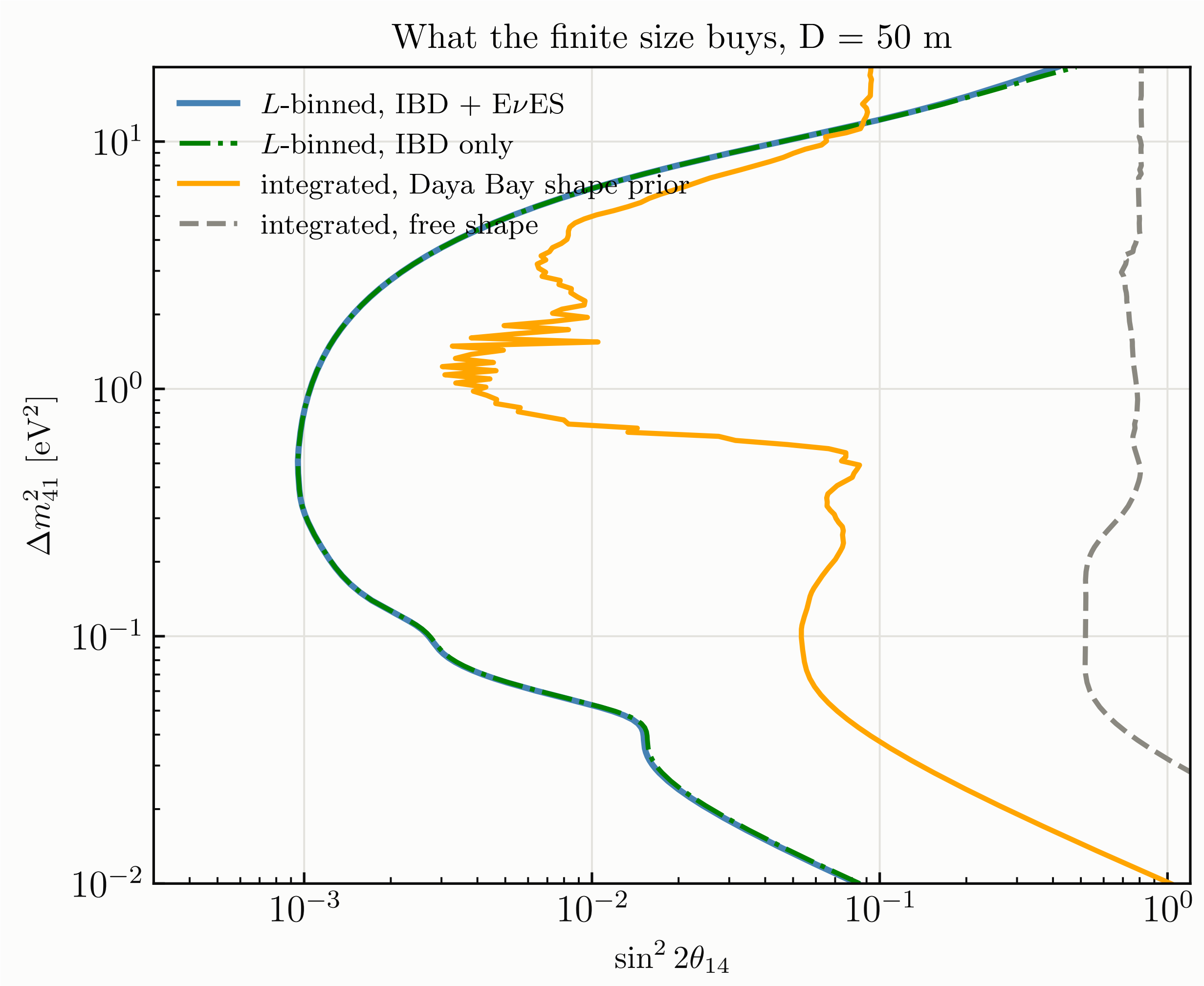

FIG. 12: The vertex-binned search against the same data analysed as a single integrated baseline, and the IBD-only variant. Two integrated curves are shown: with the same free E-shape as the vertex-binned fit, which leaves it no handle at all, and with the Daya Bay shape covariance imposed as a prior.

its surviving E-wiggles resist the flux covariance, and by up to ~ 90 elsewhere. Above $\sim 15\,\text{eV}^2$ the ordering reverses, since the L-wiggle is smeared out while the E-wiggle is not. So the finite detector is not an improvement on the conventional analysis; it is what makes a prior-free search possible at all.

- Each standoff owns roughly a decade of Δm^2 (Table III). The compact standoffs are where the sensitivity is: 500 m reaches 2×10^{-2} and 1400 m only 1.8×10^{-1}, the $1/D^2$ rate cost outrunning the extra phase. The full 3+1 formula matters at the few-percent level in the limit at 1.4 km for $\Delta m^2 \lesssim 0.1\,\text{eV}^2$ — the Δ_{34} term dilutes the wiggle, and the $(1 - s_{14})^2$

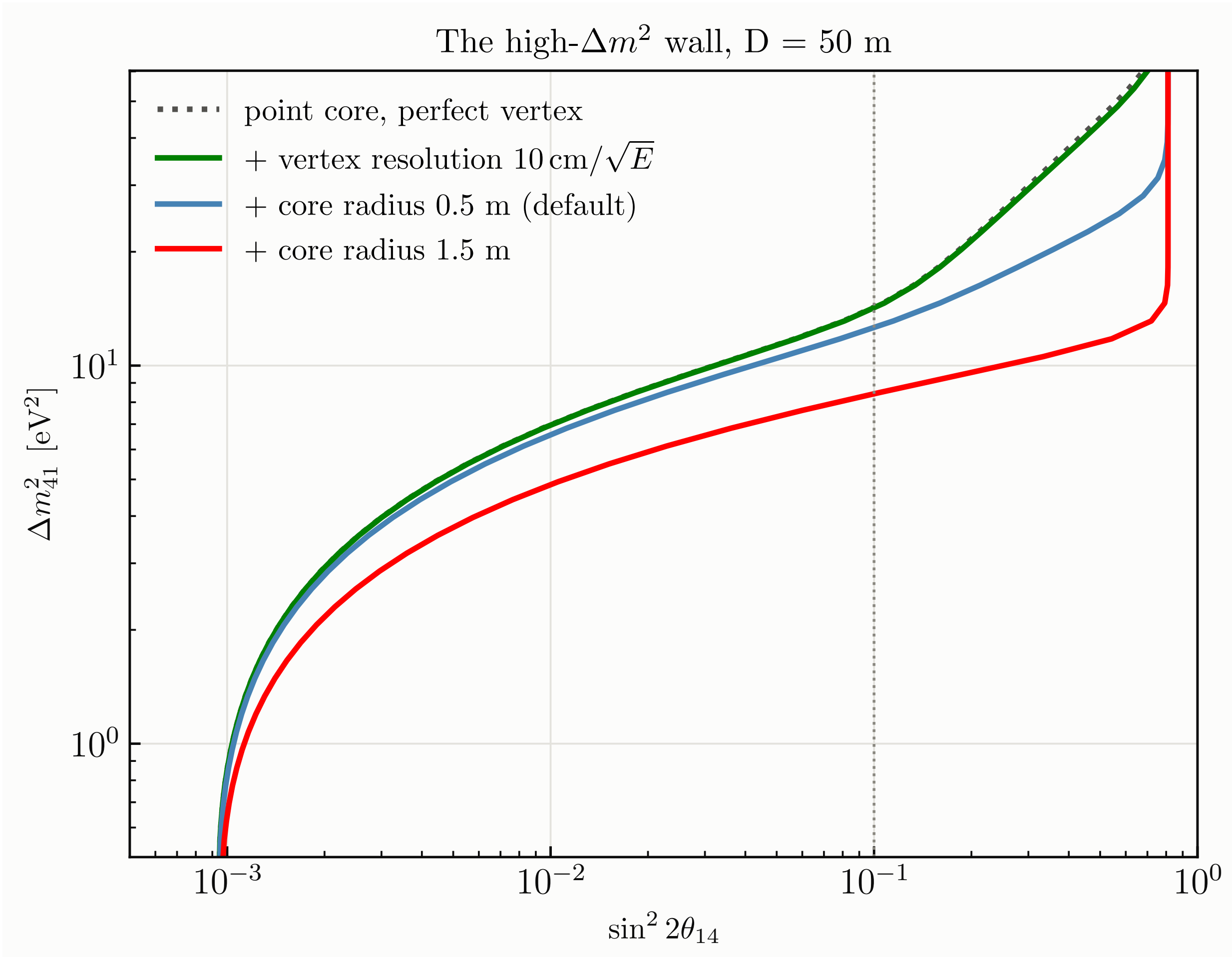

FIG. 13: The high-Δm^2 wall dissected: idealised point source and perfect vertexing, then the vertex resolution, then finite cores of 0.5 and 1.5 m. The dotted vertical line is $\sin^2 2\theta_{14} = 0.1$. Every configuration loses sensitivity above its wall rather than saturating, because a fully averaged depletion is degenerate with the free normalisation.

term of Eq. (38) adds $\sim 3\%$ — and is identically null at 50 m.

- The high-Δm^2 wall is the *energy* resolution, through the L/E phase ($\sigma_\phi \propto \Delta m^2 L \sigma_E / E^2$). Quoting the Δm^2 at which the reach crosses $\sin^2 2\theta_{14} = 0.1$: an idealised point source with perfect vertexing gives up at $14\,\mathrm{eV}^2$, JUNO's vertexing costs almost nothing on top of that (13), and the core size is the secondary effect — $12\,\mathrm{eV}^2$ for a 0.5 m core, 8 for 1.5 m. Compactness helps, but the energy resolution sets the ceiling. Above the wall every configuration loses sensitivity entirely, as it must: a fully averaged depletion is L- and E-independent and is therefore exactly the normalisation this analysis leaves free.
- What limits the reach is statistics and the shape-only construction, not the detector. Freeing

	$D = 50$ m	100 m	200 m	500 m	1400 m
IBD rate [day^{-1}]	6.1×10^4	1.5×10^4	3.7×10^3	590	71
baseline span [m]	34–66	84–116	184–216	484–516	1384–1416
best $\sin^2 2\theta_{14}$ (95%)	9.5×10^{-4}	2.1×10^{-3}	5.0×10^{-3}	2.2×10^{-2}	1.8×10^{-1}
at Δm^2_{41} [eV^2]	0.50	0.38	0.29	0.16	0.065
within 30% over	0.20–1.5	0.17–0.87	0.14–0.55	0.08–0.30	0.03–0.12

TABLE III: Sterile reach per standoff, 27 MW yr each. Each standoff owns roughly a decade of Δm^2 (the reach follows the accumulated phase $\Delta_{41} \propto \Delta m^2 D/E$) while the rate falls as $1/D^2$, so the compact standoffs carry the sensitivity and the kilometre one buys only reach in Δm^2.

the reactor E-shape costs a factor 2.4 against a pure-statistics fit — that is the price of not using the Daya Bay measurement, and it is worth paying — while the two response-uniformity fields together cost under 1%. EνES rides along but adds little: without E_ν reconstruction its wiggles wash out (the true-T ratio varies by 0.027 peak-to-peak against 0.10 in the input).

IV. AXION-LIKE PARTICLES FROM THE CORE (NOTEBOOK 7)

A. Physics and formalism

The core is a photon source of $\sim 5 \times 10^{18}\,\gamma$/s: photons scattering in the fuel convert to ALPs by Primakoff ($g_{a\gamma\gamma}$, coherent Z^2 on uranium) or Compton-like (g_{aee}) processes, stream unimpeded through shielding, and reach JUNO, where they scatter back into visible photons or electrons through the inverse processes or decay in flight. I follow the calculation of Ref. [36] for liquid-scintillator detectors, with the underlying formulas of Refs. [37, 38]; every expression below is the one implemented in `reactor/alps.py`. The Lagrangian is $\mathcal{L} \supset -\frac{1}{4} g_{a\gamma\gamma}\, a F_{\mu\nu}\tilde{F}^{\mu\nu} - i g_{aee}\, a\, \bar{e}\gamma_5 e$, and one coupling is switched on at a time.

a. Reactor photon flux. The prompt continuum is the FRJ-1 power law [37, 39]

$$\frac{d\Phi_\gamma}{dE_\gamma} = 5.8 \times 10^{17} \Big(\frac{P}{\mathrm{MW}}\Big) e^{-1.1 E_\gamma/\mathrm{MeV}}\ \mathrm{MeV}^{-1}\mathrm{s}^{-1}, \tag{44}$$

which integrates to $1.75 \times 10^{18}\,\gamma$/s above 1 MeV at 10 MW (the analytic $5.8 \times 10^{18} e^{-1.1}/1.1$). MINER's simulated flux is $\sim 20\times$ larger, so Eq. (44) is the conservative choice, and the one made by the reference analyses.

b. Primakoff production and inverse-Primakoff detection. For $\gamma(p_1)+N(p_2)\to a(k_1)+N(k_2)$ on a nucleus of mass M, charge Z, the exact $2\to 2$ cross section [37, 40] is

$$\frac{d\sigma_P}{dt} = 2\alpha Z^2 F^2(t)\, g_{a\gamma\gamma}^2 \,\frac{M^4}{t^2(M^2-s)^2(t-4M^2)^2}\Big\{m_a^2 t\,(M^2+s) - m_a^4 M^2 - t\big[(M^2-s)^2+st\big]\Big\}, \tag{45}$$

integrated over $t\in[t_{\min},t_{\max}]$, $t_{\max\,/\,\min} = m_a^4/4s - (p_{1\mathrm{cm}}\mp k_{1\mathrm{cm}})^2$ with $p_{1\mathrm{cm}} = (s-M^2)/2\sqrt{s}$, $k_{1\mathrm{cm}} = [(s+m_a^2-M^2)^2/4s - m_a^2]^{1/2}$, $s = M^2+2ME_\gamma$, on a logarithmic t-grid (the integrand is sharply forward-peaked). The atomic screening form factor is the Thomas–Fermi–Molière form $F^2(t) = [a_0^2|t|/(1+a_0^2|t|)]^2$, $a_0 = 184.15\,e^{-1/2}/(Z^{1/3}m_e)$, essential below $m_a\sim 10$ keV; the nuclear form factor is negligible for $E_a\lesssim 10$ MeV. Because $|t|/M^2\ll 1$ the ALP carries the photon energy, $E_a\simeq E_\gamma$, so the emitted flux is

$$\frac{d\Phi_a}{dE_a} = \frac{\sigma_P(E_a)}{\sigma_{\mathrm{SM}}(E_a)}\,\frac{d\Phi_\gamma}{dE_\gamma}\Big|_{E_\gamma=E_a}, \tag{46}$$

where σ_{SM} (Rayleigh + Compton + photoelectric + pair) is the total photon cross section in uranium from XCOM [41]; the branching ratio $\sigma_P/(\sigma_P+\sigma_{\mathrm{SM}})$ is $\sigma_P/\sigma_{\mathrm{SM}}$ to excellent accuracy in the coupling range of interest. Detection by inverse Primakoff on carbon and hydrogen uses $\sigma_P^{\mathrm{det}} = 2\sigma_P$ (spin counting: a spin-0 initial state instead of spin-1).

c. Compton-like production and inverse-Compton detection. For $\gamma e\to ae$ with $s = m_e^2 + 2E_\gamma m_e$, threshold $s > (m_a+m_e)^2$ i.e. $E_\gamma^{\mathrm{th}} = m_a + m_a^2/2m_e$, and $x = 1 - E_a/E_\gamma + m_a^2/2E_\gamma m_e$ [37, 42]:

$$\frac{d\sigma_C}{dx} = \frac{\alpha g_{aee}^2\, x}{4(s-m_e^2)(1-x)}\Big[x - \frac{2m_a^2 s}{(s-m_e^2)^2} + \frac{2m_a^2}{(s-m_e^2)^2}\Big(\frac{m_e^2}{1-x}+\frac{m_a^2}{x}\Big)\Big], \tag{47}$$

integrated over $x_{\min,\max} = \big[(s-m_e^2)(s-m_e^2+m_a^2)\mp(s-m_e^2)\sqrt{(s-m_e^2+m_a^2)^2-4sm_a^2}\big]/2s(s-m_e^2)$; the ALP flux is the convolution of $\sigma_C^{-1}d\sigma_C/dE_a\times\sigma_C/\sigma_{\mathrm{SM}}$ with the photon flux (Ref. [36], Eq. 14). Detection by inverse Compton $ae\to\gamma e$ uses (Ref. [36], Eq. 22)

$$\frac{d\sigma_{IC}}{dE_\gamma} = \frac{\alpha g_{aee}^2}{32\pi}\,\frac{y}{m_e^2 p_a^2 E_\gamma}\Big(1+\frac{4m_e^2E_\gamma^2}{y^2} - \frac{4m_eE_\gamma}{y} - \frac{4m_a^2 m_e p_a^2 E_\gamma \sin^2\theta}{y^3}\Big), \qquad y = 2m_eE_a + m_a^2, \tag{48}$$

integrated over $E_\gamma\in[y/2(m_e+E_a+p_a),\, y/2(m_e+E_a-p_a)]$ with $\cos\theta = (m_e+E_a-y/2E_\gamma)/p_a$. Axio-electric absorption is neglected: it scales as Z^5 and on carbon at MeV energies is $\sim 10^{-4}$ of the germanium rates of the reference analyses, negligible against inverse Compton here.

d. Decays, survival and decay-in-detector probabilities.

$$\Gamma(a\to\gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}, \qquad \Gamma(a\to e^+e^-) = \frac{g_{aee}^2 m_a}{8\pi}\sqrt{1-\frac{4m_e^2}{m_a^2}}\ \ (m_a > 2m_e), \tag{49}$$

with lab decay length $\ell_a = (p_a/m_a)\,\hbar c/\Gamma_a$. An ALP produced in the core survives to the detector centre with $P_{\rm surv} = e^{-L/\ell_a}$ (for scattering, evaluated at $L = D = 50$ m) or to the front face with $e^{-(D-R)/\ell_a}$, and then decays inside with probability

$$P_{\rm decay} = 1 - \exp\big(-\bar{\ell}/\ell_a\big), \qquad \bar{\ell} = \tfrac{4}{3}R = 22\,\mathrm{m}, \tag{50}$$

the mean chord of the fiducial sphere (this replaces the fixed depth $r = 0.7$ m of the JUNO-TAO geometry). The decay-length boundary of Ref. [36], Eq. (53), corrected to $g^2 = 8\pi\hbar c/(Lm_a\beta)$, is reproduced numerically to 10^{-4}.

e. Event yields. Over a live time Δt (27 MW yr = 3 yr at 90% duty), with $\mathcal{A} = \pi R^2 = 855\,\mathrm{m}^2$ the detector face and $N_{C,H,e}$ the fiducial target counts,

$$N^P_{\rm scat} = \frac{\Delta t}{4\pi D^2}\int dE_a\,\frac{d\Phi_a}{dE_a}\,e^{-D/\ell_a}\,2\big[N_C\sigma^C_P(E_a) + N_H\sigma^H_P(E_a)\big], \tag{51}$$

$$N^{IC}_{\rm scat} = \frac{\Delta t}{4\pi D^2}\int dE_a\,\frac{d\Phi_a}{dE_a}\,e^{-D/\ell_a}\,N_e\,\sigma_{IC}(E_a), \tag{52}$$

$$N_{\rm decay} = \frac{\mathcal{A}\,\Delta t}{4\pi D^2}\int dE_a\,\frac{d\Phi_a}{dE_a}\,e^{-(D-R)/\ell_a}\,\big[1 - e^{-\bar{\ell}/\ell_a}\big], \tag{53}$$

with E_a integrated over the single-hit window 3–10 MeV. Compared with JUNO-TAO we trade a GW reactor for 10 MW and 30 m for 50 m, but gain $\sim 2\times10^4\times$ the target mass and $\sim 25\times$ the decay path; the decay term (53) is what carries the reach.

f. Validation. I checked every ingredient. $\Gamma(a\to\gamma\gamma)$ and $\Gamma(a\to e^+e^-)$ agree with analytic evaluation to 10^{-4} and vanish below the e^+e^- threshold. The exact Primakoff form (45) agrees with an independent implementation of the Dent et al. forward/recoilless form ($d\sigma/d\cos\theta = \frac{1}{4}g^2\alpha Z^2F^2|\vec{p}_a|^4\sin^2\theta/t^2$) at the 6–40% level, with the largest deviations near threshold and at high energy, which is exactly the deviation Ref. [37] (footnote 5) reports for that approximation. The Compton cross section has the right threshold and positivity, and the right magnitude ($\alpha g^2/m_e^2$, i.e. of order Klein–Nishina at $g_{aee} = 1$: 3.3×10^{-25} against $2.1\times10^{-25}\,\mathrm{cm}^2$ at 1 MeV). The uranium $\sigma_{\rm SM}$ reproduces its XCOM anchors, and the flux integral its analytic value.

g. Background and statistics. The single-hit window is 3–10 MeV. The background is our own model of Section II: reactor EνES 3.3×10^5, IBD singles 3.6×10^5 and solar EνES 3.1×10^4 events over the exposure (7.3×10^5 in total). Cosmogenics (^{12}B and friends) are not modelled; since $\chi^2 \propto g^8/N_b$, the reach in the coupling scales only as $N_b^{1/8}$, and even a $4\times$ larger background costs 20%. Following Ref. [36], Eq. (54), the single-bin statistic is $\chi^2 = N_s^2/(N_s + N_b + \sigma^2_{\rm sys}N_s^2)$ with $\sigma_{\rm sys} = 0.1$, and 90% CL is $\Delta\chi^2 = 4.61$ (2 dof). The background-free limit is shown alongside, since the prompt/delayed photon and vertex handles of a LS detector could approach it.

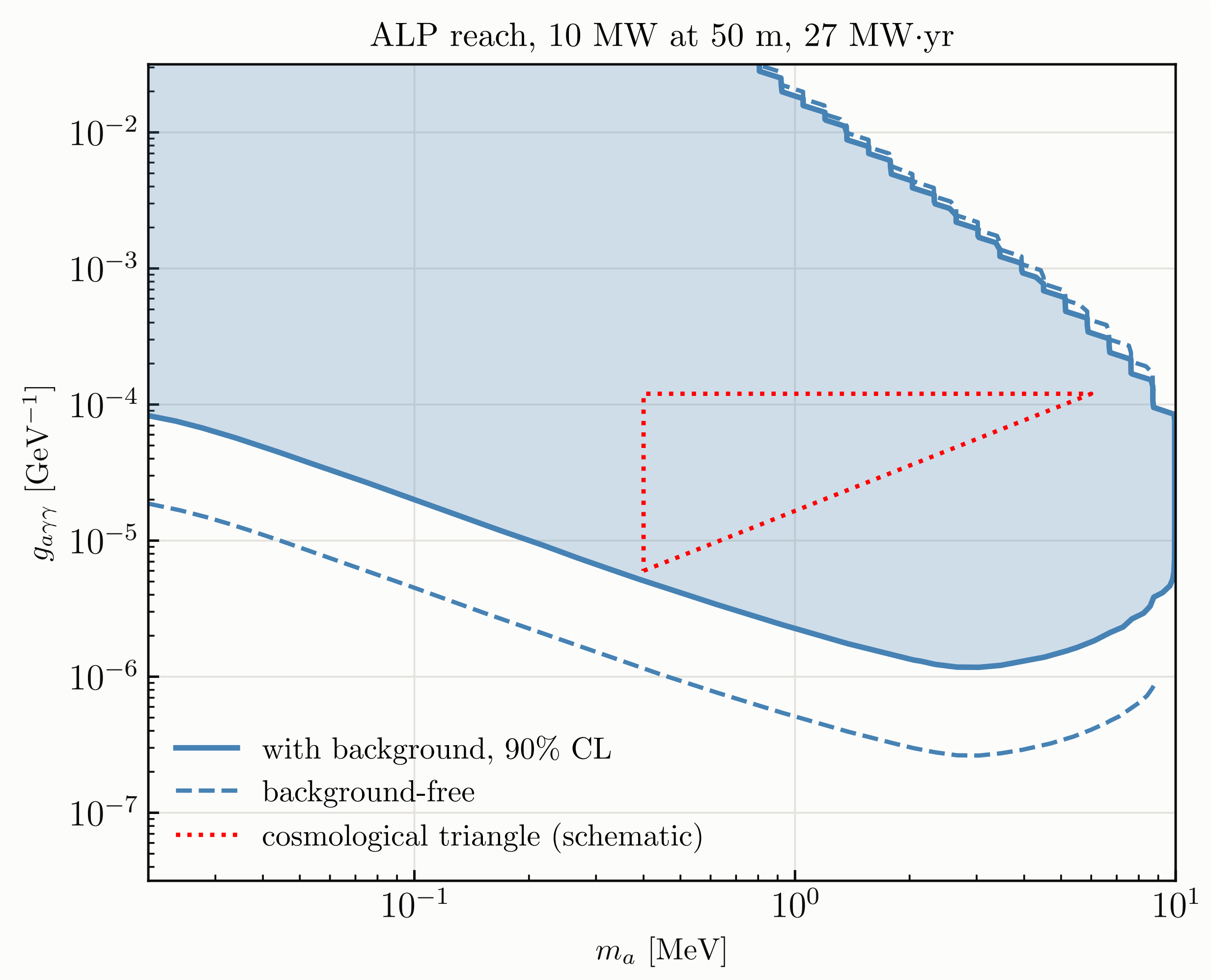

FIG. 14: 90% CL exclusion in the ALP–photon plane for 27 MW yr, with the full background (solid, filled) and background-free (dashed). The dotted wedge is the schematic cosmological triangle of Ref. [37], Fig. 10.

B. Results

The decay channel dominates by orders of magnitude at MeV masses (Fig. 14): at $m_a =$ 1 MeV and $g_{a\gamma\gamma} = 10^{-5}\,\mathrm{GeV}^{-1}$ there are 7×10^5 decays against 0.06 scatterings over the exposure. With the background included, at $m_a = 1$ MeV the excluded band is $g_{a\gamma\gamma} \in [2.3 \times 10^{-6},\ 1.4 \times 10^{-2}]\,\mathrm{GeV}^{-1}$, so the schematic cosmological triangle, the wedge at $m_a \sim 0.3$–5 MeV and $g_{a\gamma\gamma} \sim 10^{-5}$–$10^{-4}\,\mathrm{GeV}^{-1}$ between the beam-dump, HB-star and SN1987A exclusions where only cosmology-dependent bounds apply [43, 44], is covered in full. For scale, JUNO-TAO with background reaches $g \sim \text{few}\times 10^{-4}$ [36]; the near-reactor plus JUNO combination goes an order of magnitude deeper at the same reactor-off cost. In the electron plane (Fig. 15), at $m_a = 1.5$ MeV

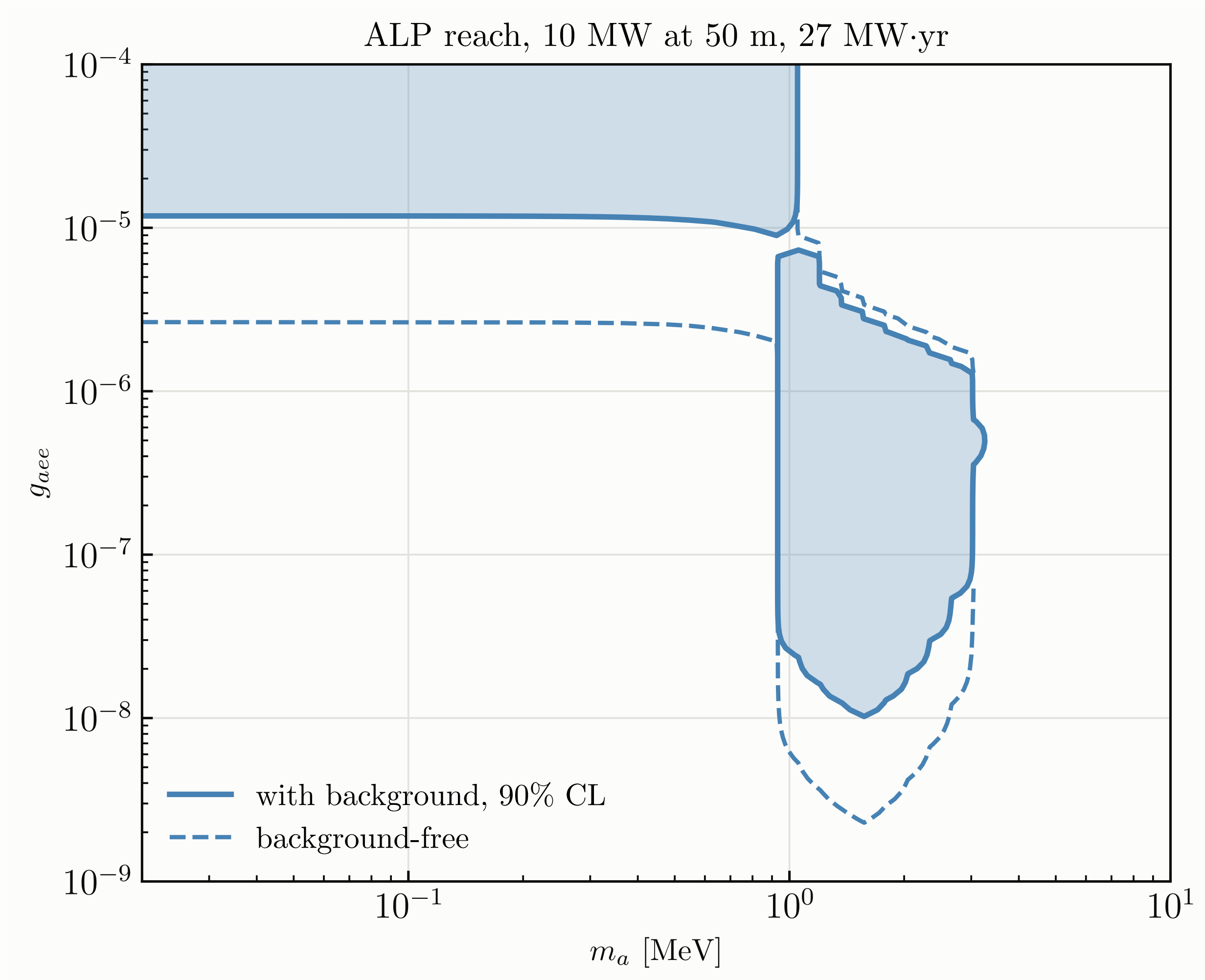

FIG. 15: Same in the ALP–electron plane; above the e^+e^- threshold the decay channel opens the band down to $g_{aee} \sim 10^{-8}$.

the excluded band is $g_{aee} \in [1.2 \times 10^{-8},\ 3.1 \times 10^{-6}]$ with background, so the unexplored stripe that Ref. [36] identifies is reached without the background-free assumption TAO needs. The caveats are the FRJ-1 flux, the absence of cosmogenics, single-bin counting (an energy-shape fit would only help), the pure-uranium core, the neglect of ALP attenuation in the shielding, and the schematic triangle overlay.

V. DISCUSSION AND CAVEATS

The three studies share a lesson about what a parked reactor's *size and proximity* do for a 20 kt detector. For the SM couplings the win is the IBD anchor: 20× the EνES statistics with the *same* flux, cancelling normalisation, flux shape and energy response against a transfer budget of a few tenths of a percent. For sterile oscillations the win is the detector's finite size, which converts the oscillation into an L-direction observable that no source systematic can mimic, and which crucially allows a shape-only analysis independent of the Daya Bay measurement that would otherwise carry the signal. For ALPs the win is the decay volume: an $855\,\mathrm{m}^2$ face and a 22 m chord watching a $5 \times 10^{18}\,\gamma$/s source.

The stated caveats are real. The detector singles are subtracted with reactor-off running (Section II B), but their spectral shape is a component model rather than a transport simulation and the cosmogenic rates are depth-scaled estimates; what protects the result is that it depends on them only through $\sqrt{B}$. They are still unmodelled in the ALP window, where they would enter the reach as $N_b^{1/8}$ and cost a few percent. The reactor spectrum is undefined below the IBD threshold, which is where 48% of the EνES window rate sits; that costs the answer nothing, but it means the quoted EνES rate is a lower bound. The transfer budget of Section II is essentially the whole of the weak-mixing-angle result, and its leading term — N_e/N_p times the relative selection efficiency, assumed to 0.5% — is not a neutrino measurement but a scintillator-composition one, which is where the effort belongs. The two response-uniformity fields of the sterile search (0.5% efficiency and 0.2% energy scale over the volume) are assumptions that should be replaced by JUNO's calibration numbers, though that search turns out not to be limited by them. A companion feasibility estimate (notebook 8) finds that the reactor's own neutron and gamma leakage require a research-reactor-class shield (~ 6 m water-equivalent, or ~ 2 m steel plus a sub-metre high-Z collar, on top of JUNO's 4 m of pool water) for the near-reactor programme not to spoil single-hit physics; this is exactly what a TRIGA-style pool installation provides. Finally, the mobile-reactor θ_{13} study (notebook 2) remains the strongest oscillation argument for such a source: the fixed-reactor variant (notebook 2b) shows that the 33 m detector provides only 0.026 rad of θ_{13} phase and cannot substitute for moving the source. The same finite size that is the right tool at metre oscillation lengths is the wrong one at kilometres.

Appendix A: Daya Bay flux model and covariance

a. The measured spectra. Daya Bay's unfolding [3] releases the per-fission IBD yields Φ_i^0 [$\mathrm{cm}^2\,\mathrm{fission}^{-1}$] in $n = 25$ neutrino-energy bins of width 0.25 MeV over $E_\nu \in [1.8, 8.0]$ MeV, for three spectra (^{235}U, ^{239}Pu, and the "Total", their cycle-averaged mixture at fission fractions 0.564 : 0.076 : 0.304 : 0.056), with a 75×75 covariance V over the stacked [^{235}U, ^{239}Pu, Total] vector. The near-reactor studies of this report use the ^{235}U yields with the 25×25 block, which is the only measured shape uncertainty they carry: the ^{239}Pu component is 6.3% of fissions at mid-cycle and 12.5% at end of cycle, and its shape freedom is represented by the ±30% fuel-evolution mode of Eq. (3) rather than by its own covariance block. The standard JUNO analysis (below) uses the Total.

Figure 16 shows the measured ^{235}U and ^{239}Pu yields with their released 1σ uncertainties (the diagonal of V; the last bin spans 7.8–9.5 MeV), converted into IBD rates at JUNO for a hypothetical pure 10 MW core of each isotope at 50 m, together with the Huber–Mueller ^{238}U and ^{241}Pu spectra used for the minority components. The Huber–Mueller polynomial fits are valid only up to 8.5 MeV; above it the code applies an artificial exponential cutoff (marked in the figure), which is why the ^{238}U and ^{241}Pu curves kink there. This is immaterial, since these are 1–2% components and contribute nothing to any analysis above 8 MeV. Figure 17 is the companion: the actual HALEU core at the start ($\beta = 0$), middle and end ($\beta = 1$) of its cycle, built from those ingredients through Eqs. (1) and (2), with a 5% rate drop and a slight softening ($2.00 \to 1.90 \times 10^7$ IBD per year); its bottom panel shows the shape change as a ratio to the start of cycle. Figure 18 is the same figure for a commercial-LEU core, the Daya Bay measured fission fractions across their cycle ($F_{235} = 0.66 \to 0.49$, $F_{239} = 0.23 \to 0.36$), at the same power and distance. The LEU core has a 5% lower absolute rate (plutonium-rich fuel yields fewer $\bar{\nu}_e$ per unit power), and because the HALEU model applies Daya Bay's Pu *increments* on top of a ^{235}U-dominated fresh core, its end-of-cycle distortion ($\sim 10\%$ at 8 MeV) is essentially the LEU one. That is the statement "follows the Daya Bay trajectory" made visible, and the reason it is a conservative model for a 2%-^{238}U core, whose real plutonium breeding is slower.

b. From binned yields to a continuous flux (the NuFit prescription). Following Appendix A of the NuFit analysis that the repository's standard method reproduces, the continuous flux times cross section is written as a smooth reference shape times a modulation,

$$S(E) = \phi_{\mathrm{HM}}(E)\, \sigma_{\mathrm{IBD}}(E) \sum_{m=1}^{n} y_m\, \delta_m(E), \tag{A1}$$

where ϕ_{HM} is the Huber–Mueller spectrum at the Daya Bay average fission fractions (with no isotope correction), σ_{IBD} is Vogel–Beacom, and $\{\delta_m\}$ are the cardinal cubic splines through the 25 bin centres ($\delta_m(E_{m'}) = \delta_{mm'}$). The coefficients are fixed by demanding that the bin-averaged prediction reproduce the measured yields exactly:

$$M_{im} = \frac{1}{\Delta E_i} \int_{\mathrm{bin\,i}} \phi_{\mathrm{HM}} \sigma_{\mathrm{IBD}}\, \delta_m\, dE, \qquad \boldsymbol{y}^0 = M^{-1} \boldsymbol{\Phi}^0, \tag{A2}$$

with the integrals done on a dedicated 1 keV grid; the residual $\max_i |(M\boldsymbol{y}^0)_i - \Phi_i^0|/\Phi_i^0$ is 6×10^{-16}. Above the last measured bin (9.5 MeV after the release's extension) the modulation is held constant.

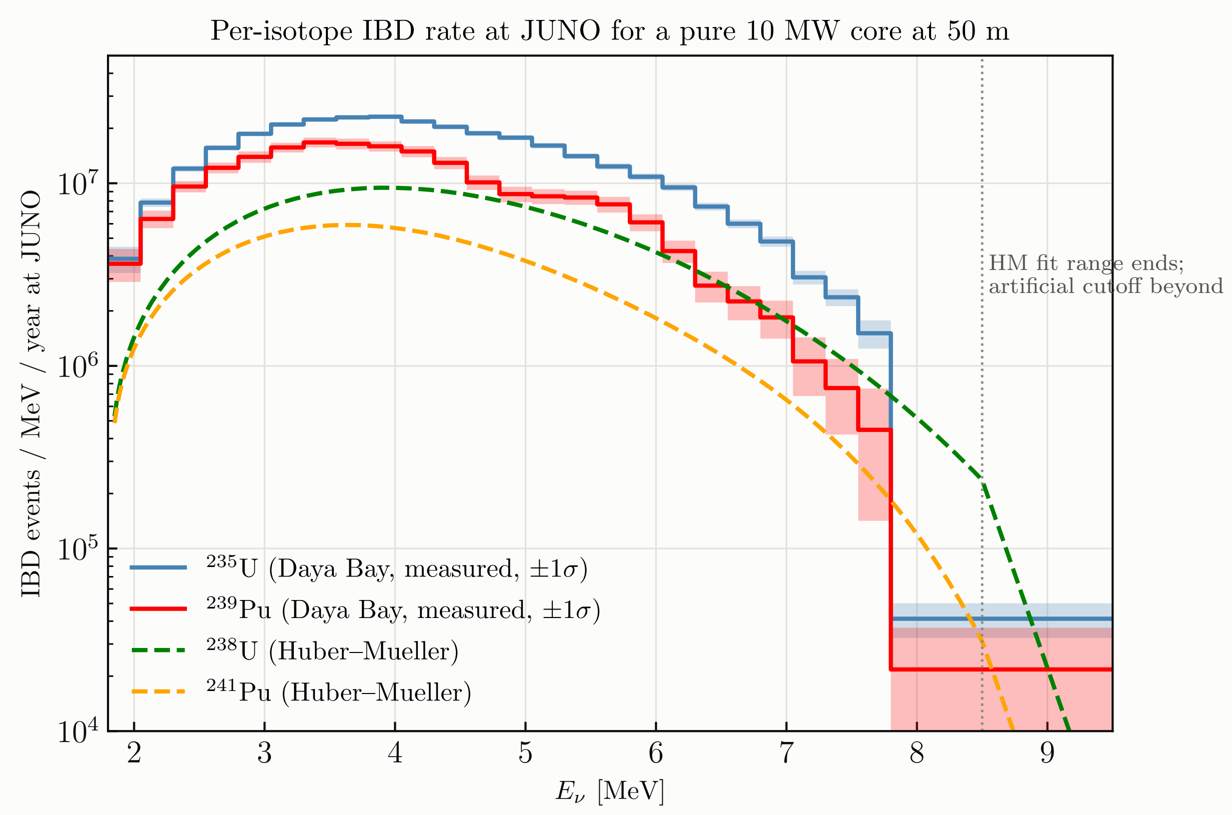

FIG. 16: Per-isotope IBD rate at JUNO for a pure 10 MW core of each isotope at 50 m: the measured Daya Bay ^{235}U and ^{239}Pu yields [3] with their released 1σ uncertainties as bands, and the Huber–Mueller [5, 6] ^{238}U and ^{241}Pu spectra (dashed). The dotted line marks the end of the Huber–Mueller fit range.

c. From covariance to pulls. The covariance is recast as n unit-Gaussian pulls by eigendecomposition, $V = O\,D\,O^T$, $\Psi = O\sqrt{D}$ (columns are modes), and each mode is pushed through the same interpolation, $\boldsymbol{y}^{(k)} = M^{-1}\Psi_{\cdot k}$, so that a pull ξ_k distorts the flux by

$$S(E) \;\to\; S(E)\Big[1 + \sum_k \xi_k \,\frac{\sum_n y_n^{(k)}\delta_n(E)}{\sum_n y_n^0\delta_n(E)}\Big], \qquad \chi^2 \supset \sum_k \xi_k^2, \tag{A3}$$

which is exact to first order and keeps the pull penalty diagonal. In the near-reactor modules the same construction is applied to the relative covariance $V_{ij}/(\Phi_i^0\Phi_j^0)$ of the ^{235}U block, and in the joint fit of Section II each of the 25 modes acts on the IBD anchor and the EνES spectrum *coherently*, which is what lets the anchor measure it away; the modes are then interpolated linearly in E and enter the Asimov covariance $C = \mathrm{diag}(\mu) + \sum_k v_k v_k^T$ as $v_k = t\,\partial\mu/\partial\xi_k$. Modes with eigenvalue below 10^{-10} of the largest are dropped. In the sterile analysis of Section III this prior is deliberately *not* used, and the flux shape and normalisation are left free, for the reason explained

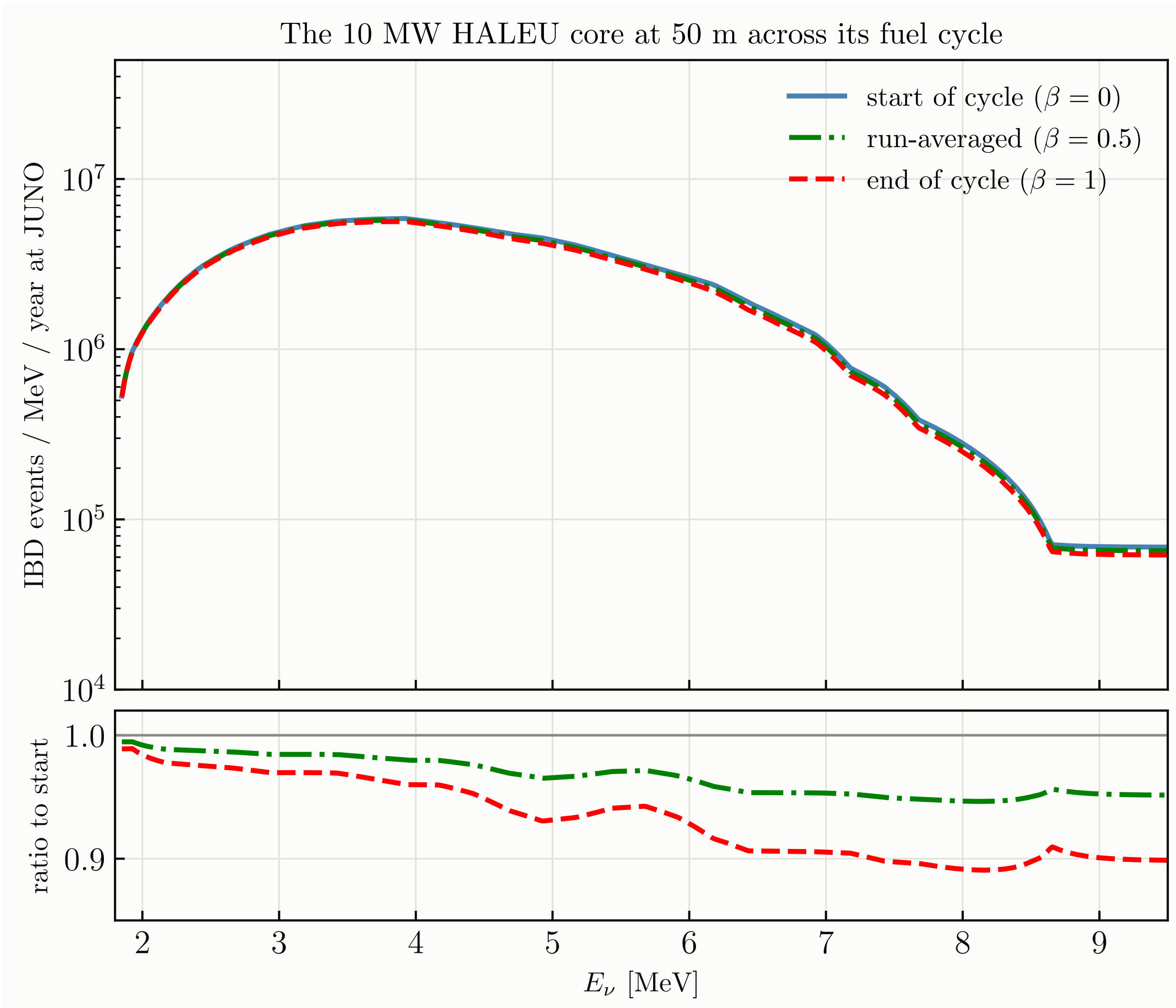

FIG. 17: The 10 MW HALEU core at 50 m at start, middle and end of its fuel cycle (Eq. (2)), built from the spectra of Fig. 16; bottom: ratio to the start of cycle.

there.

d. The measured Total flux and JUNO's own spectrum. For the standard far-reactor analysis, the NuFit prescription additionally rescales the prediction bin-per-bin to JUNO's recovered un-oscillated spectrum,

$$k_i = \frac{\left(N_i^{\text{obs}} - B_i\right)/\overline{P}_{ee,i}}{N_i^{\text{pred,unosc}}}, \tag{A4}$$

where $\overline{P}_{ee,i}$ is JUNO's own released model survival probability per bin; this pins the un-oscillated shape to data and turns flux-model and cross-section choices into nulls (it was found to be the decisive ingredient in reproducing JUNO's solar-parameter result to 0.06σ). It has no analogue for projections, where the measured flux with its covariance takes its place.

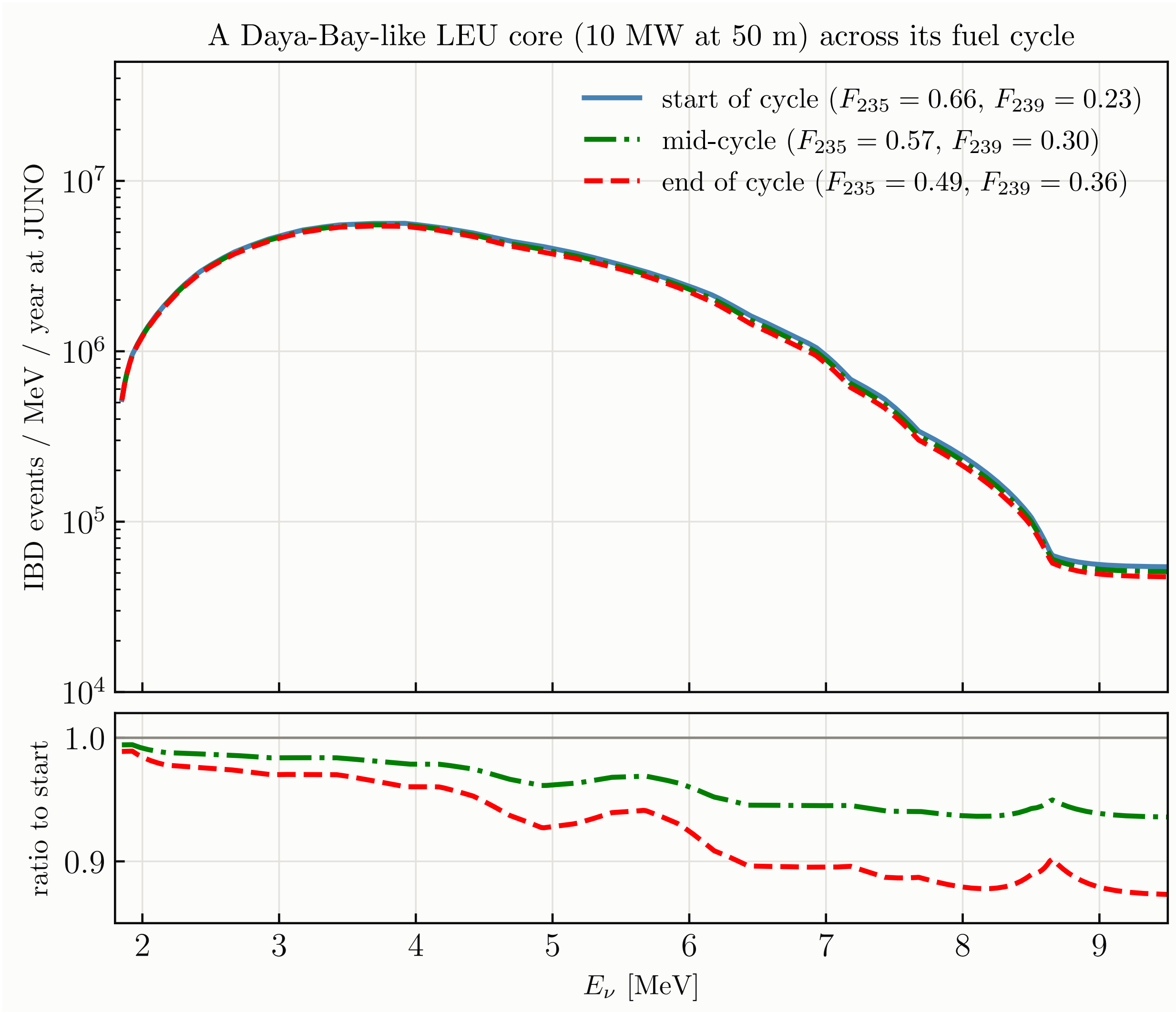

FIG. 18: Same for a commercial-LEU core: the Daya Bay measured fission fractions [2] at the start, middle and end of their 20 burnup groups, at the same 10 MW and 50 m so the two figures are directly comparable.

Appendix B: Systematics of the standard JUNO analysis (NuFit procedure)

The repository's standard far-reactor analysis reproduces the NuFit 6.1 treatment of the JUNO 59.1-day data [7], and its nuisance set is what the near-reactor studies inherit for the far side and for the detector response. The test statistic is

$$\chi^2 = \sum_i \frac{\left(N_i^{\text{obs}} - N_i^{\text{pred}}(\boldsymbol{\xi})\right)^2}{\sigma^2_{\text{CNP},i}} + \sum_a \left(\frac{\xi_a}{\sigma_a}\right)^2, \qquad \frac{3}{\sigma^2_{\text{CNP},i}} = \frac{1}{N_i^{\text{obs}}} + \frac{2}{N_i^{\text{pred}}}, \tag{B1}$$

the combined Neyman–Pearson form used by JUNO (NuFit's own default is the Poisson Baker–Cousins form $2\sum_i[P_i - O_i + O_i \ln(O_i/P_i)]$; the two agree to 10^{-4} in $\sin^2\theta_{12}$). The prediction

pull	meaning	prior σ
$\xi_{\rm norm}$	signal normalisation	2.4% = 1.8% (Tab. 2 rates) $\oplus$ 1.6% (selection efficiency)
$\xi_{\rm scl}$	energy scale	0.5%
$\xi_{\rm bias}$	energy bias (offset in $F_{\rm nl}$ units)	0.5%
$\xi_{\rm res}$	resolution	5%
$\xi_{\rm LiHe}$	^{9}Li/^{8}He normalisation	33%
$\xi_{\rm sh}$	^{9}Li/^{8}He linear-in-E shape (20% at 1 MeV)	0.20
$\xi_{\rm geo}$	geoneutrino normalisation	42%
$\xi_{\rm world}$	world-reactor normalisation	10%
$\xi_{\rm BiPo}$	^{214}Bi–^{214}Po normalisation	56%
$\xi_{\rm other}$	other backgrounds	100%
ξ_k, $k = 1..25$	Daya Bay flux-shape modes (Section A)	1 each
$r_{\rm BG}$	overall background rescaling (all but geoneutrinos)	fixed, 1.00 (NuFit cnf 2: 1.15)
$r_{\rm nl}$, $r_{\rm res}$	non-linearity / resolution rescalings	fixed, 1.0 (cnf 3: 1.024; cnf 5: 1.3)

TABLE IV: Nuisance parameters of the standard analysis. The 2.4% normalisation is the NuFit "note added" budget; their published Tab. 1 configurations use 1.8%. External inputs: $\sin^2\theta_{13} = 0.02175$, $\Delta m^2_{ee} = 2.466 \times 10^{-3}\,\text{eV}^2$ (Daya Bay).

is

$$N_i^{\rm pred} = (1{+}\xi_{\rm norm})\, k_i\, \mathcal{N} \sum_{\rm cores\,c} \Big[R\big(\xi_{\rm scl}, \xi_{\rm bias}, \xi_{\rm res}\big)\; S\big(E; \{\xi_k\}\big)\, \frac{P_{ee}(E, L_c)}{4\pi L_c^2}\Big]_i {+} \sum_b B_{b,i}\,(1{+}\xi_b)\,(1{+}\delta_{b,\rm LiHe}\,\xi_{\rm sh} E_i), \tag{B2}$$

with the nine signal cores (Yangjiang, Taishan and the effective Daya Bay core at 215 km), matter effects at 2.55 g cm^{-3}, and the response

$$\tilde{E}_{\rm pr} = E_{\rm pr}\, r_{\rm nl}\big[(1+\xi_{\rm scl})F_{\rm nl}(E_{\rm pr}) + \xi_{\rm bias}\big], \qquad \sigma(\tilde{E}) = (1+\xi_{\rm res})\, r_{\rm res} \sqrt{a^2\tilde{E} + b^2\tilde{E}^2}, \tag{B3}$$

$F_{\rm nl}$ being JUNO's released positron non-linearity and $a = 3.3\%$, $b = 1.0\%$. Table IV lists every pull and its prior in the standard configuration.

a. Profiling. The prediction is affine in the three energy pulls (through the response matrix) and the 25 flux pulls (through the spectrum) at fixed oscillation parameters, so a linear basis is built once per grid point and the 28 pulls plus the background nuisances are profiled analytically; the two oscillation parameters are minimised numerically. In the Asimov projections of Sections II and III the same nuisances become the joint modes of $C = \mathrm{diag}(\mu) + \sum_k v_k v_k^T$ and no minimisation is performed at all.

b. What the reproduction established. With this treatment the repository reproduces NuFit's Tab. 2 relative fluxes exactly, their best fit to 0.10σ, and their Δm^2_{ee} profile; with the note-added budget it lands on JUNO's official $(\sin^2\theta_{12}, \Delta m^2_{21})$ at 0.06σ with every nuisance well inside its prior. The decisive ingredients, in order, were including the Daya Bay complex in the signal (its $\Delta_{21} \approx 2\pi$ position fills the solar dip), the bin-per-bin rescaling of Eq. (A4), and the 2.4% normalisation. The residual difference in the ellipse tilt corresponds to about 3% of effective rate freedom, that is, only $\sqrt{3^2 - 2.4^2} \approx 1.8\%$ beyond the documented budget, and it must be coherent in energy (candidates: the delivered-power history, the absolute Daya Bay flux anchor, the target proton count, and off-equilibrium/spent-fuel corrections).

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